

Web Clipping Developer's Guide

Document Number 3009-001
Print Date 11/3/99

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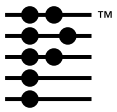
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1



Introduction

Web Clipping Developer's Guide shows you how to develop Web content for the Palm VII™ connected organizer. If you can create Web pages and construct simple Web sites and know how a Palm™ organizer works, you're already halfway there.

This book provides guidelines on designing information flow, designing your "site," and using HTML to good effect. It is vitally important that you read this small book in its entirety before attempting to develop Palm Query Applications (PQAs) and Web clippings (usually referred to as "clippings"). The similarities with Web pages are obvious, but the crucial differences in goals and technology are not quite so apparent.

This chapter provides an overview of PQAs, Web clippings, and the expertise and tools needed to develop them for the Palm VII organizer. It also includes an outline of the life-cycle of a PQA.

Chapter 2, "Design Guidelines," is essential reading for the Web clipping developer. Be sure to read its discussions of PQA design goals, the process of gathering, sorting, and partitioning information, information and file structure, and interface design considerations. This chapter also includes notes on using advertising in a PQA.

Chapter 3, "PQA and Clipping Interface Elements," gets down to the nuts and bolts of constructing the HTML source files of the PQA. Consult this chapter after reading Chapter 2. Technical information on page and file size, directory structure, hyperlinks, font usage, form elements, and graphics is provided. Special attention is given to the use of special META tags and other elements.

Chapter 4, "Security and Authentication," provides a brief discussion about the built-in security features of the Palm VII organizer. There is also practical advice on how to ensure client authentication over the air.

Chapter 5, "Building and Testing PQAs," is an essentially procedural chapter covering the use of the Query Application Builder, the Image Checker, and the Palm OS™ Emulator in compiling and testing PQAs.

See "HTML and Palm OS Extensions for PQAs" and "Palm OS HTML 3.2 DTD" for complete information on supported HTML elements, attributes, and extensions.

What Is a PQA?

A Palm Query Application is a kind of mini-Web site that resides on the Palm VII organizer. Technically, it's a special kind of record database. A typical PQA contains an HTML form or a list of hyperlinks that request additional information either locally—on the handheld computer—or remotely—on the Internet. A PQA can also be self-contained, a sort of HTML booklet with internal cross-references, formatting, and graphics. You can also launch other Palm OS™ applications from a PQA.

Figure 1.1 Typical PQA on a Palm VII Organizer



You author a PQA in HTML—complete with hyperlinks and graphics—and compile it in the Query Application Builder. The user then installs the compiled `.pqa` file to the Palm VII organizer via HotSync software (or by beaming from another Palm device). By default, the PQA is displayed in the Applications Launcher in the Palm.Net™ category. When tapped, the PQA is rendered by the Web Clipping application, an application otherwise invisible to the user.

What Is a Web Clipping?

A Web clipping is a results page—an HTML page that answers the query sent from the PQA on the Palm VII organizer over the air to an Internet server. Clippings are typically small pages generated by

a CGI script from the user's query, though they can be static pages on an Internet server as well.

Figure 1.2 Typical Web Clipping Page



Expertise

To effectively create PQAs and clippings, you should have the ability to write straightforward HTML (including links, graphics, tables, and form elements). You should be able to write or specify CGI scripts. And you should have a good sense of how to design economical and easy-to-use pages. (This book does not discuss authoring CGI scripts.)

PQAs and clippings are authored in a subset of HTML 3.2. The HTML 3.2 specification is available from the World Wide Web Consortium at http://www.w3c.org/TR/REC-html32.html. See also "Overview of the Palm.Net System" in *Palm OS Programmer's Companion*.

For good discussions and practical guidance on designing Web pages, see Patrick J. Lynch and Sarah Horton, *Web Style Guide: Basic*

Design Principles for Creating Web Sites (New Haven and London: Yale University Press, 1999); an on-line version of this book is available at http://info.med.yale.edu/caim/manual.

Most importantly, you should have a good understanding of how to design information flow. This includes partitioning information between the organizer-side PQA and the server-side clipping—and conceptualizing the links within the PQA, from the PQA to other Palm™ applications, and from the PQA to the Internet.

Developer Support

There are several sources of on-line support for PQA developers:

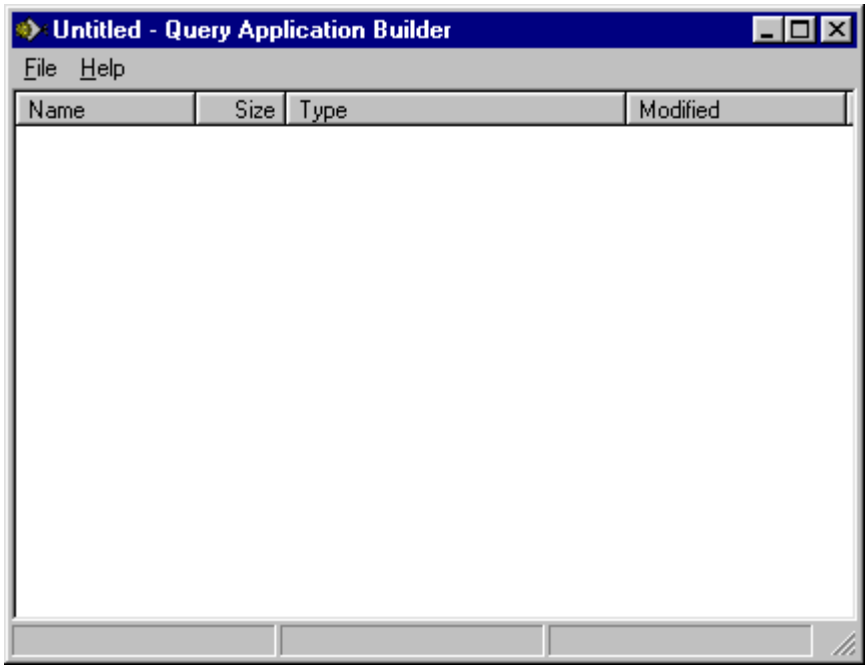
- White papers and other documentation can be found at http://www.palm.com/devzone/palmvii/.
- You can join the PQA developer forum at http://www.palm.com/devzone/mailinglists.html.
- If you have questions for Developer Support, see http://www.palm.com/devzone/questions.html.
- Additional PQAs for downloading and comparison can be found at http://www.palm.net/apps/index.html.

Tools

You need only a few tools to develop PQAs and clippings:

- A text editor or non-WYSIWYG HTML editor. WYSIWYG HTML editors often insert either nonstandard or incompatible elements that cannot be transmitted to or rendered on the Palm VII organizer.
- The Query Application Builder, which runs on Windows 95, Windows 98, Windows NT, and Mac OS (System 7.x–8.x). You use the Query Application Builder to compress and compile PQAs. You can download the Query Application Builder from http://www.palm.com/devzone/palmvii/.

Figure 1.3 Query Application Builder



- The Palm Image Checker graphics viewer. Although this software is not required, you can use the Image Checker to verify the display of your PQA graphics. The Image Checker is included in the Windows Query Application builder archive; the Image Checker is built in to the Mac OS Query Application Builder executable.
- Palm Desktop software for the Palm VII organizer, including HotSync® manager (to install the PQA).

The Palm OS Emulator (POSE) is necessary if you want to create and test your PQAs on the desktop computer without a Palm VII organizer. You can download POSE from http://www.palm.com/devzone/pose/.

Alternatively, you can create PQAs in the Query Application Builder on the desktop computer and install them on a Palm VII organizer for testing.

See Chapter 5, "Building and Testing PQAs," for information on the use of the Query Application Builder and Image Checker.

Webs for the Wireless

The mechanics of authoring PQA and clipping pages differ very little from the mechanics of authoring Web pages and CGI scripts or Java servlets in general. However, because of the constraints of authoring for a handheld organizer connected to the Internet via a relatively expensive radio link, you will need to rethink your philosophy of design and implementation.

Much of the content on the Internet is designed to take advantage of the power of Pentium/RISC-class computers with large, high-resolution color monitors and fast and cheap Internet access. In these circumstances, there is little reason to economize on the abundant connect time and large file size that make Web browsing such a rich, multimedia experience from a desktop or notebook computer.

The world looks different when you're holding a small, low-power computer like the Palm VII organizer with its tiny screen, battery-powered operation, and relatively slow and expensive wireless connection to the Internet. Rather than try to duplicate the Web-browsing model on a handheld computer, you should strive to develop PQAs that access targeted bits of Internet information—like taking clippings from a newspaper.

Typically, Palm VII users do not focus on following hyperlinks to the Internet (though these remain a key part of the PQA); instead, they compose off line a simple query in the PQA (for example, a request for a stock quote) and then send that query over the air.

The radio-transmitted query is received by a local base station, which relays it to the Palm™ Web Clipping proxy servers (often referred to as "proxy servers") of the Palm.Net service. From there, the query is sent on to the appropriate Internet address. The HTML results page—the clipping—retraces these steps to arrive at the user's Palm VII organizer.

When developing PQAs and clippings, remember that the Basic Palm.Net plan allows for the transmission of 50 KB per month; the Expanded plan allows for 150 KB per month. Every byte up and down counts!

In the PQA, users can also access locally stored information—HTML pages and graphics. And developers can store graphics and pages in the PQA for later use in downloaded clippings.

Life Cycle of a PQA and a Web Clipping

You create a PQA in a four-step procedure:

1. You author HTML pages and graphics to serve as the base for the PQA. (Use designers and editors, if they are available to your project!)
2. You convert the HTML pages and graphics in the Query Application Builder to produce the PQA file.
3. You test the PQA either in POSE (2.1d26 or later) or on a Palm VII organizer or both. (It's also a very good idea to submit the PQA to usability testing!)
4. You distribute the compiled `.pqa` file and users install it with HotSync software on their Palm VII organizers. You can also beam a PQA from one Palm organizer to another.

Development of a Web clipping is very similar:

1. You design the HTML template, optimizing it for over-the-air transmission to and display on the Palm VII organizer. This book tells you how to optimize your HTML files for this purpose.
2. You author the server-side scripts to generate HTML clipping pages when appropriate queries are received.

Of course, you can also place static HTML pages on the Internet server for transmission as clippings. Generally speaking, though, you will want to keep static information in the PQA on the Palm VII organizer and dynamically generate clippings on the Internet server. See Chapter 2, "Design Guidelines," for advice on partitioning information between PQA and clipping pages.

From the user's perspective, interaction with a PQA couldn't be easier:

1. The user installs the PQA with HotSync software as if it were any other Palm application. By default, the installed PQA is displayed in the Palm.Net category of the Applications Launcher.

2. The user taps the PQA icon to open the application.
3. The user fills out a form or taps a link to a locally stored page or graphic or to request a clipping from an Internet server. The requested information is displayed on the user's Palm VII organizer.

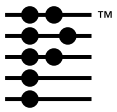
The process is a little more complicated under the hood, however, with interactions between user, Applications Launcher, Web Clipping application, PQA, Web Clipping proxy server, and Internet server.

Once the user launches the PQA and taps a link—

1. If the linked page or graphic is installed on the Palm VII organizer, the page or graphic is displayed. Unbeknownst to the user, it's the Web Clipping application that does the rendering of the compiled HTML pages and graphics. (The user never explicitly invokes the Web Clipping application—the pages "just appear.")
2. Otherwise, the Palm VII organizer compresses the query into a small file in Palm Query Format and sends this packet over the air to the local base station (part of the BellSouth wireless network).
3. The local base station relays the compressed packet to the Web Clipping proxy servers on the Palm.Net service.
4. The Web Clipping proxy server translates the query packet into standard HTTP packet and then sends the decompressed query to the appropriate Internet address.
5. The HTTP server returns an HTML clipping to the Web Clipping proxy server.
6. The Web Clipping proxy server compresses the HTML clipping into a Palm Proxy Format file and sends it back to the local base station.
7. The local base station relays the clipping to the Palm VII organizer, where the Web Clipping application renders the page for viewing.

Communication between the Palm VII organizer and the Web Clipping proxy server is via the User Datagram Protocol (UDP)(to quickly send a query packet up and a response packet down). Communication between the Web Clipping proxy servers and Internet servers is via standard Internet protocols: TCP, HTTP, SSL.

2



Design Guidelines

Just as a Palm VII connected organizer is not a miniaturized desktop computer but a compact machine designed for specific purposes, the PQA is not just a miniaturized Web site. The Palm organizer and the PQA are smaller—and they are also different.

Like other Palm OS applications, PQAs are meant to provide information-on-demand to the mobile user.

PQA Design Goals

Before designing, laying out, and linking HTML files and graphics for your PQA—and much before compiling them with the Query Application Builder!—you should take the time to think your query application through.

Put the same thought and care into your PQA that you would put into a multimedia Web site. Before proceeding, address the questions usually related to site development:

- What are the goals of your PQA?
- Who is the user of the PQA? What does this user want and expect from your application? How expert is the typical user?
- What kind of information do you want to convey with your PQA?
- How does this PQA reflect your corporate or organizational identity?

Meet with your development team—or, if you are the one-person development team, meet with other interested colleagues—and produce at least a memorandum outlining your answers to these questions. In the heat of coding, such a written statement will come in very handy as a point of reference.

Example Goals Memo

We want TheNews PQA to do three things very well:

- Provide on-demand news headlines for local news (wherever the user happens to be in the U.S.A.)
- Provide, as a next step in querying, brief news stories related to the headlines
- Focus the user on other company resources and services (which may be delivered via PQA or over the Web to a personal computer)

The user of TheNews PQA is typically a manager on the run between meetings, conferences, hotels, airports—on the go and a frequent traveler. Highly educated, high news consumer (CNN in the hotel room, subscriber to a local paper and *The New York Times* at home). User is in need of local news—as a habit but also as "cultural" preparation for social meetings in diverse locales. "I hear there's a new highway coming through downtown!"

We see the user requesting headlines and then selecting a few stories to download and read. The stories would be limited to a paragraph (usually, the first paragraph in an inverted-pyramid story). A limited number of news pictures would also be available, but the user would have to explicitly request them—per picture or in preferences setting. Otherwise, the cost might not justify the convenience.

Our company specializes in repackaging and delivering information from diverse sources to the people who need it—managers and executives in general but targeted by industry segment, especially. TheNews PQA is a low-overhead way of providing some information while publicizing our other, more focused on-line services.

Gathering and Sorting Information

The main purpose of any PQA is to provide the timely and accurate information that the Palm VII user desires. Logically, an important early step in design is the collection of the information you intend your PQA to deliver. This is true no matter the kind of information: airline schedules, wine lists, news headlines, what have you.

Collect the information and then organize it into useful categories—for instance, you could sort a list of wines by country, region, varietal, vintner, year, quality, or color. Where possible, arrange the

information in a simple hierarchy based on importance to the user. Your goal is the design of a simple database, even if it consists of only a stack of index cards.

You're now beginning to see the shape of the content your PQA is meant to deliver. Let the shape of the content drive the design of the pages and the logic of the hyperlinks. (Designing pages first is a sure way to paint yourself into a corner.)

File Structure Considerations

The file structure of your PQA is the one place where the shape of the content is not necessarily reflected. Your information can be arranged hierarchically, but your files must all reside under the same root directory. So, for example, the home page of your PQA is, in terms of information hierarchy, superior to the local and remote pages linked to it. But all the local pages in the PQA are treated as files residing in the same directory. See "PQA Directory Structure" on page 27 for more technical information on file and directory structure.

Partitioning Your Information

Because the Web clipping scheme requires some information to reside on the Palm VII organizer and other information to reside on Internet servers, you have to make a cut somewhere—to partition the information in a sensible way. This is a crucial step in designing the information flow of your PQA.

To begin with, more stable information should be included in the PQA and installed on the Palm VII organizer. The query form, graphics files, and frequently consulted but nonvolatile information should typically be made part of the PQA.

More volatile information should be stored on the Internet server for downloading and display only as requested on the Palm VII organizer.

Once you have partitioned your information, you are ready to design the details of information flow—and then go on to design and lay out PQA and clippings pages.

Designing Information Flow

Information flow—how kinds and chunks of information are linked and in what order they are presented—should always be viewed from the user's point of view. Start with the user's needs and expectations and work backward from there.

In addition to users' needs, you should base information flow on the constraints and purposes of a handheld computer accessing the Internet over an expensive, low-bandwidth radio connection:

- Implement only absolutely required features, and make sure that the most frequently used features are the most easily accessible ones.
- Help users formulate precise queries so that the results downloaded from the Internet are small in size and to the point.
- Use local—on-device—hyperlinks to present additional information, when possible.
- Present only the most essential information to the user and resort to over-the-air hyperlinks only when absolutely necessary.
- When possible, store graphics for clippings locally in the PQA, rather than transmit them over the air.
- Remember that whatever you store on the Palm VII organizer will remain relatively static (of course, users can update PQAs with HotSync software or beam a PQA from one Palm organizer to another).
- Speed counts. Users will notice even a one-half-second delay.

PQA Structure

Every PQA should have a top-level or "index" page. This is the first page displayed when the user launches the PQA on the Palm VII organizer. Hyperlinks to other HTML pages and graphics in the PQA and on the Internet start here.

Single-page PQAs

A PQA could easily contain only one on-device page: a query form with its bare minimum of supporting graphics (for example, the large and small icons displayed in the Applications Launcher). A user would compose a query—using text boxes, check boxes, pop-up menus, etc.—and then submit the completed form for over-the-air transmission to download a results page. The results page could supply all the information the user needed.

In this case, your PQA structure is very simple. The index page is a form that requests one page for downloading and display. For example, a user might want to look up the telephone number of the three nearest towing services: a simple request generating a simple answer.

Figure 2.1 A One-page PQA (Except for Help and Legal Pages)



In another possible format for a one-page PQA, you could use the page to list Internet hyperlinks. Clicking a link would initiate an over-the-air transaction to download a clipping on an Internet server.

Multipage PQAs

Often, though, PQA structure will be a bit more complicated, with multiple pages and graphics on the handheld computer. And results pages could link to local pages and graphics as well as to Palm OS applications and remote hyperlinks. You need to give a shape to the relationships between PQA pages, clippings, and graphics—before you start laying them out and authoring them.

In most cases, the information hierarchy of PQAs with multiple pages should be shallow. Users will not want to drill down through too many layers to find that bit of information they think is important. Otherwise, why would they look for it in the first place? All computer users are impatient—users of wireless handhelds have even less patience than when they are browsing the Web through their desktop computers.

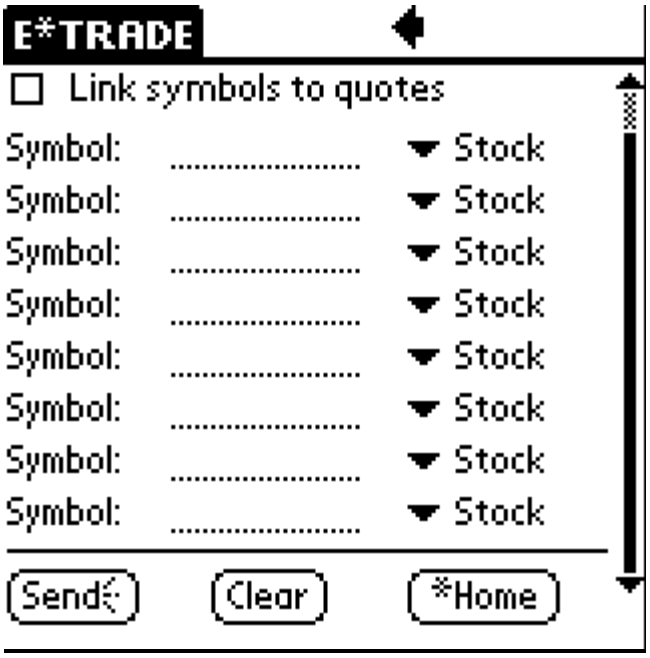
Figure 2.2 Multipage PQA



Once users drill down, they will want to make their way back up the information hierarchy—either to a previously visited page or to the top-level index page. The Web Clipping application

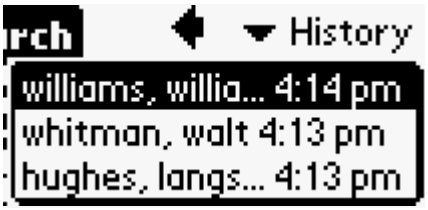
automatically supplies a back-arrow button. But consider supplying home buttons and other links to make navigation more intuitive.

Figure 2.3 Home Button Example



The History list is a useful way for the user to access cached clippings. And each clipping can include a special META tag to provide a custom string that helps identify the clipping in the History list. See "Basic Structural Elements" on page 28 for information on using the HistoryListText META tag.

Figure 2.4 Using the History List



In any case, strive for ease of navigation and let the shape of the content determine the relationships between the pages.

A Note on Using Hyperlinks

Just as the information you want to convey is partitioned between handheld computer and Internet server, your links must also be partitioned. You will want to make use of local links to navigate between pages and images in the PQA and between the PQA and other Palm OS applications. See "Hyperlinks" on page 36 for technical details.

Help users navigate by maintaining their sense of context: similarity in pages create a feeling of predictability. But remain aware of the difference between exploratory Web browsing and PQA use.

You will, of course, make remote links to content on the Internet. Again, the look and feel of the results pages should match that of the query pages—with the proviso that users readily grasp the difference between remote and local pages and links.

Interface Design Considerations

Strive to fulfill the basic tenets of interface design: clarity, consistency, and utility. Good interface design is even more important when developing content for the small, gray-scale screen of the Palm VII organizer.

Going beyond these obvious goals of user interface design, keep in mind the following:

Consistent Look and Feel

All pages should have a consistent look and feel to enable users to easily navigate between pages and across applications. Strive for recognition—so that users don't have to relearn the interface. Whenever possible, design elements should be similar to those used in other Palm OS applications. Make sure your pages are well organized and easy to read. Avoid excess verbiage and keep functionality simple. Say and show only what users need.

User Control, Query Formation, and Results

Users pay for every byte transferred over the air. Plan your application in such a way as to keep connect time to a minimum.

Leave it up to users to initiate all wireless connections. Make sure that users understand when an action causes a wireless transaction. Clearly label commands and dialog boxes to indicate which ones will send and receive data over the air.

The Palm OS supplies over-the-air icons that are displayed when a remote hyperlink or button is rendered. One over-the-air icon is for HTTP links and the other is for secure HTTPS links. But these icons are not automatically displayed in graphical links. In this case, you should include an over-the-air icon in the graphic itself. Use your graphics software to copy and paste the Palm icons.

Figure 2.5 Over-the-air Icon



Figure 2.6 Secure Over-the-air Icon



Give users the tools they need to make more precise queries. For example, your PQA could include data such as airport codes or ticker symbols. It could also include a small glossary of terms as used on your site.

Your PQA can display multiple pages before a clipping is requested and retrieved. But you should limit the number of layers, since the point of the PQA is to help users access critical information quickly. Consider using pop-up menus to help users form their queries.

Figure 2.7 Pop-up Menu Example



If the content of a clipping is typically larger than 500 bytes, let users know. You could do this by providing a visual cue, such as text in a button that indicates the average size of a download (e.g., “Full Story ~2K”). You could also put this information in the help file of the PQA.

The Palm VII title bar displays double exclamation marks (!!) when a clipping larger than 1 KB (compressed) is received. This tells users that the download was more expensive than usual. In any case, when users tap the title area, the full title of the clipping (if the clipping author included text in the <TITLE> element) and number of bytes received are displayed.

Advertising

The PQA is an excellent place to provide company information, showcase additional products and services, and build brand loyalty.

Advertisements should not be obtrusive—be parsimonious with fonts and graphics.

Advertisements can appear both in the query form and in the clipping. Store small ad banners in the PQA file so that query forms and clippings can easily and quickly reference them.

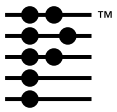
A Western reader's eye tends to move from the upper-left corner of the screen to the lower-right corner. Consider placing your company's logo or an icon in the header.

Figure 2.8 Company Logo in PQA



You can also send graphics over the air, but this adds to user expense. Avoid wireless hyperlinks in ads, unless the link is to a Palm friendly site. See "Basic Structural Elements" on page 28 for information on the PalmComputingPlatform META tag.

3



PQA and Clipping Interface Elements

This chapter highlights the differences between standard HTML 3.2 and the subset of tags you can use in authoring PQA and clipping pages. Palm OS extensions to HTML 3.2 are discussed and examples given. You will also find indications of how to use HTML within the constraints of the Palm VII organizer.

The technical requirements of PQA and clipping pages (e.g., sizes, obligatory META tags, treatment of graphics) are covered as are recommendations for reducing the size of pages and over-the-air queries. Chapter 1, "Introduction," and Chapter 2, "Design Guidelines," are *essential* reading before you get down to coding your HTML files for PQAs and clippings.

See Chapter 4, "Security and Authentication," for information on ensuring client-side security and authentication. See Chapter 5, "Building and Testing PQAs," for information on using the Query Application Builder.

See "HTML and Palm OS Extensions for PQAs" and "Palm OS HTML 3.2 DTD" for complete information on supported HTML elements, attributes, and extensions.

Page Length, Number of Pages, File Size

There is no technical limitation on the length of a PQA or clipping page, but it's best to limit the amount of scrolling a user has to do. Try and design your pages to display on one screen.

No single, compressed PQA page can exceed 63 KB—this is a system memory limitation. But this would be a very, very large file for a Palm VII organizer! Practically speaking, the size of the total, compiled PQA—all pages and graphics—should average less than 15 KB. You can build as many pages as you like into your PQA. The

absolute technical limitation on PQA size is determined only by system memory.

The Query Application Builder compresses HTML files by 5–60 percent. The compression ratio varies according to just what kinds of files are included in your PQA. But when you compile a PQA, the Query Application Builder shows you how much compression occurs.

The size of the query sent up to the Web Clipping proxy server should be kept down to about 40 bytes.

Clippings sent down from the Web Clipping proxy server should be as small as possible—about 360 bytes (compressed). Again, since the compression ratio varies, you need to test your clippings pages over the air to determine their compressed size.

Minimizing Size of PQAs, Clippings, and Queries

Given that every single byte is precious everywhere in Web clipping technology—in system memory, in queries sent up, and in results pages sent down—take pains to minimize the size of your pages and queries.

An obvious way to reduce the size of PQA and clipping pages is to minimize the use of space-hungry graphics. But every text character counts as well. This includes HTML markup: use HTML markup only when necessary.

Similarly, you can reduce the size of queries sent over the air by using short, lowercase variable names in form input elements. You can also work to reduce the number of characters the user needs to input. And, if you can, require the user to input only lowercase characters.

For example, instead of requiring the user to write the name of a state (e.g., “New York”), provide a pop-up list of postal codes (e.g., the user picks “ny” from the menu). The string associated with the OPTION element becomes part of your over-the-air query. Keep such strings as short as possible. See “Query Form Elements” on page 39 for more on pop-up menus.

PQA Directory Structure

You should organize your PQA around a home page, traditionally titled index.html. This page should provide the query form or the principal local and Internet links to more specific content.

The Query Application Builder requires that all PQA source files—HTML pages and graphics—be under a single root directory: subdirectories are permitted for the purposes of building a PQA. Hyperlinks between source pages and graphics to be built in to the PQA should be relative paths that include subdirectories. However, clippings must reference local pages and graphics as if those pages and graphics were stored in the same flat directory. This is because the Query Application Builder turns HTML and graphics files into Palm database records in the PQA.

For example, in an HTML file that is to be compiled into the PQA, you could reference a local file from any page in the PQA with a link like this:

```
<a href="subdirectory/page.html">textstring<a>
```

But to reference this page from a clipping, you would use a link like this (note that there is no subdirectory in the link):

```
<a href="file:mypqa.pqa/page.html">textstring<a>
```

See also “Hyperlinks” on page 36.

Content Area

The content area for your HTML page is 153x144 pixels. Using the normal Palm font (TD 9), only eleven lines can be displayed at a time, including the header or banner. See “Fonts” on page 31.

The Navigation bar is located at the top of each page, making this area unavailable for other content.

When laying out pages, always go to the edge of the screen—do not waste space on borders.

HTML for PQAs and Clippings

PQA and clippings pages are authored in a subset of HTML 3.2. Several now-common Web page features are not supported in PQAs and clippings, including

- Named typefaces
- Style sheets
- Image maps
- Frames
- Nested tables
- Scripts and applets
- Cookies

However, you can use simple tables, gray-scale color, limited font markup, lists, and images, as discussed below. And there a few Palm tags that optimize HTML for PQA and clipping pages.

See "HTML and Palm OS Extensions for PQAs" and "Palm OS HTML 3.2 DTD" for complete information on supported HTML elements, attributes, and extensions.

Basic Structural Elements

The standard HTML elements that structure an HTML page are, of course, supported: <HTML>, <HEAD>, <TITLE>, and <BODY>. These elements are not required for PQA and clippings pages, but their use is strongly encouraged. See Listing 3.1 and Listing 3.2 for typical and required tags for PQA and clipping pages.

IMPORTANT: The text you specify with <TITLE> is displayed in the Navigation bar. Keep this text short and test to make sure it fits on screen.

There are three Palm-specific META tags that you should use:

- PalmComputingPlatform – Identifies PQA or clipping page as Palm Friendly

- HistoryListText – Stores clipping page name in History list
- PalmLauncherRevision – Sets the version string for your PQA

Another Palm META tag, LocalIcon, is used to store images and graphics in a PQA for use by a clipping page. Use of LocalIcon is covered in "Hyperlinks" on page 36.

PalmComputingPlatform

Syntax:

```
<meta name="palmcomputingplatform" content="true">
```

Use PalmComputingPlatform in all PQA and clipping pages to indicate that they were authored for a Palm VII organizer. This ensures that images smaller than 153 pixels are included and that your PQA builds properly.

IMPORTANT: Clipping pages without the PalmComputingPlatform tag are truncated by the proxy server to 1 KB and images are removed. PQA pages without this tag build incorrectly: images you reference in the HTML pages comprising the PQA will not be included in the PQA file.

HistoryListText

Syntax:

```
<meta name="historylisttext" content="yourstring
&date &time">
```

Use HistoryListText in all clipping pages to store them in a list of clippings in the History list cache. You can use &date and &time variables to append the current date and time. The string you supply for CONTENT is included in the History list. (If you don't use this tag, the name of the PQA is displayed, instead.)

PalmLauncherRevision

Syntax:

```
<meta name="palmlauncherrevision" content="x">
```

x is a string representing the version number of your PQA; e.g., 1.0, 2.1.

Include PalmLauncherRevision in the index page of your PQA. This META tag sets the version string of your PQA, which is displayed in the Info window of the Applications Launcher. Use of this META tag makes it much easier for developer and user to determine when an upgrade is necessary.

Listing 3.1 Skeleton PQA Page

```
<html>
<head>
<title>Sample Page for PQA</title>
<meta name="palmcomputingplatform" content="true">
<meta name="palmlauncherrevision" content="1.0">
<meta name="localicon" content="logo1.gif">
<meta name="localicon" content="logo2.gif">
</head>
<body>
<!-- your text and links here -->
</body>
</html>
```

Listing 3.2 Skeleton Clipping Page

```
<html>
<head>
<title>Sample Clipping</title>
<meta name="palmcomputingplatform" content="true">
<meta name="historylisttext" content="Latest News - &date &time">
</head>
<body>
<!-- your text and links here -->
</body>
</html>
```

Using the Same Markup for Palm VII Screen and Web Browser

The Web Clipping application recognizes a Palm extension to HTML that makes it possible for you to use the same HTML pages for clippings and normal Web browsing. You can hide any clipping content that you don't want displayed on the Palm VII screen by surrounding that content with the tags, <SMALLSCREENIGNORE></SMALLSCREENIGNORE>.

If you do use the same HTML pages—whether static or CGI generated—for Web browsing and Web clipping, you should make the pages Palm Friendly in other respects.

Listing 3.3 Skeleton Clipping Page with <SMALLSCREENIGNORE>

```
<html>
<head>
<title>Sample Clipping with Undisplayable Content</title>
<meta name="palmcomputingplatform" content="true">
<meta name="historylisttext" content="Latest News - &date &time">
</head>
<body>
<!-- your text and links here -->
<smallscreenignore>
<!-- an 8-bit, full-screen graphic that the Palm VII screen
cannot display-->
<img src="montblanc.jpg" alt="Mont Blanc">
</smallscreenignore>
</body>
</html>
```

Fonts

Remember that large fonts take up additional screen real estate—use them sparingly. Likewise, limit your markup of in-line font changes—italic, boldface, etc.—to only necessary emphasis and editorial requirements. Palm TD is the only typeface supported by

the Web Clipping application. TD 9 is the normal font for plain text; this corresponds to <FONT SIZE=3>.

You can change font sizes with the <SMALL>, <BIG>, and <FONT SIZE=*n*> tags. You can also use the <BASEFONT SIZE=*n*> tag. The effects of some of the size markup may be surprising. But underlying this scheme is the limitation on the font sizes that the Web Clipping application can display.

Table 3.1 and Table 3.2 show how HTML markup is rendered on the Palm VII organizer. The normal font for each tag is shown as are the effects of using <SMALL> and <BIG> to change the size. The Smallest and Biggest columns show the range of sizes available in the <FONT SIZE=*n*> element.

Table 3.1 HTML Headings and Fonts

HTML	Normal	<small> Font	<big> Font	Smallest <font>	Biggest <font>
<h1>	TD 12 B	TD 11 B	TD 12 B	TD 11 B	TD 12 B
<h2>	TD 11 B	TD 10 B	TD 12 B	TD 9 B	TD 12 B
<h3>	TD 10 B	TD 9 B	TD 11 B	TD 8 B	TD 12 B
<h4>	TD 8 B	TD 8 B	TD 10 B	TD 7 B	TD 11 B
<h5>	TD 8 B	TD 8 B	TD 9 B	TD 7 B	TD 10 B
<h6>	TD 7 B	TD 7 B	TD 8 B	TD 7 B	TD 9 B

Table 3.2 HTML Text Elements and Fonts

HTML	Normal	<small> Font	<big> Font	Smallest <font>	Biggest <font>
<address>	TD 9 B It	TD 8 B	TD 10 B	TD 7 B	TD 11 B
<b>	TD 8 B	TD 8 B	TD 10 B	TD 7 B	TD 11 B
<blockquote>	TD 9	TD 8 B	TD 9 B	TD 7 B	TD 11 B

Table 3.2 HTML Text Elements and Fonts (*continued*)

HTML	Normal	<small> Font	<big> Font	Smallest <font>	Biggest <font>
<cite>	TD 9 It	TD 8 B	TD 9 It	TD 7 B	TD 10 B
<dfn>	TD 9 It	TD 8 B	TD 9 It	TD 7 B	TD 10 B
<dir>	TD 9	TD 8 B	TD 9 B	TD 7 B	TD 11 B
<dl>	TD 9	TD 8 B	TD 9 B	TD 7 B	TD 11 B
	TD 9 It	TD 8 B	TD 9 It	TD 7 B	TD 10 B
<i>	TD 9 It	TD 8 B	TD 9 It	TD 7 B	TD 10 B
<kbd>	TD Mono	TD Mono	TD Mono	TD Mono	TD Mono
<listing>	TD Mono	TD Mono	TD Mono	TD Mono	TD Mono
<menu>	TD 9	TD 8 B	TD 9 B	TD 7 B	TD 11 B
	TD 9	TD 8 B	TD 9 B	TD 7 B	TD 11 B
<p>	TD 9	TD 8 B	TD 9 B	TD 7 B	TD 11 B
<plaintext>	TD Mono	TD Mono	TD Mono	TD Mono	TD Mono
<pre>	TD Mono	TD Mono	TD Mono	TD Mono	TD Mono
<sample>	TD Mono	TD Mono	TD Mono	TD Mono	TD Mono
	TD 8 B	TD 8 B	TD 10 B	TD 7 B	TD 11 B
<table>	TD 9	TD 8 B	TD 9 B	TD 7 B	TD 11 B
<tt>	TD Mono	TD Mono	TD Mono	TD Mono	TD Mono
	TD 9	TD 8 B	TD 9 B	TD 7 B	TD 11 B

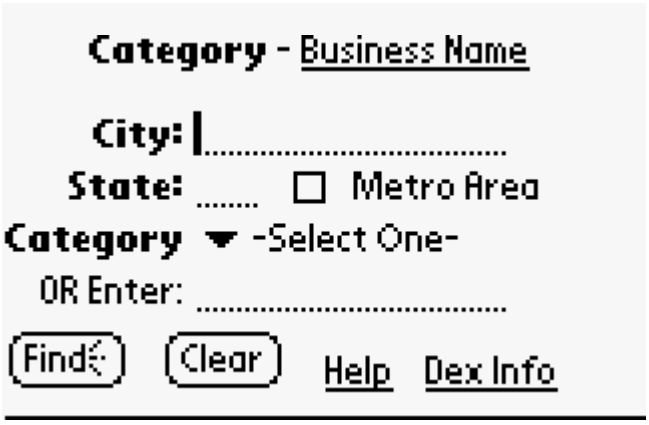
Table 3.2 HTML Text Elements and Fonts (*continued*)

HTML	Normal	<small> Font	<big> Font	Smallest <font>	Biggest <font>
<var>	TD Mono	TD Mono	TD Mono	TD Mono	TD Mono
<xmp>	TD Mono	TD Mono	TD Mono	TD Mono	TD Mono

Editorial and Design Tips

Use boldface for labels and normal type for data items.

Figure 3.1 Laying Out Labels and Data



The meaning of labels should be immediately clear to users. Don't use boldface extensively—it's best for contrast, but hard to read as a text type.

Use italics to show emphasis and to mark book titles, foreign words, etc. (Refer to a good editorial style guide for full treatment of when and what to italicize.) Avoid using underline—since hyperlinks are normally underlined, anyway, use of this in-line markup will cause confusion.

You will only rarely want to use strikethrough (<S> or <STRIKE>) or monospace type in your PQAs and clippings. Keep your pages clean and easy to read—and quick to transmit and load.

Most of the basic HTML elements—for example, heads (<H1>–<H6>), horizontal rules (<HR>), line breaks (
), preformatted text (<PRE>), block quotes (<BLOCKQUOTE>), address (<ADDRESS>), ordered or numbered (<OL>) lists, unordered or bulleted (<UL>) lists, and definition lists (<DL>)—are available.

Gray-scale Colors

The Palm VII organizer supports these four gray-scale colors:

- black (#000000)
- gray (#808080)(rendered as dark gray)
- silver (#C0C0C0)(rendered as light gray)
- white (#FFFFFF)

The default colors are black text on white background. You can specify the other supported colors with the BGCOLOR and TEXT attributes in the BODY element. You can also specify colors with the COLOR attribute of the FONT and BASEFONT elements.

These colors are predefined, so you can use the color name (e.g., gray) instead of the hexadecimal representation of the color (e.g., #808080).

Tables

Use simple tables to display stock portfolios, timetables, or lists of restaurants or hotels. You can also use tables to lay out and divide PQA and clipping pages.

Since table wrapping is not supported, a table wider than 153 pixels will be truncated.

Nested tables—tables within tables—are not supported in PQA or clipping pages. If you try to include a nested table, its content will be rendered as if it were content of the parent table cell that contains the nested table.

Although the WIDTH attribute of the TABLE element is supported, values represented as percentages are not. You can provide pixel values, instead. However, it is best to leave the sizing of tables to the Web Clipping application, which will use the “best fit” algorithm to format the table to fit the Palm VII screen.

The WIDTH attribute of the TD (table cell) element is also supported. In this case, you *can* provide percentage values. When no WIDTH attribute is given, the Web Clipping application applies the "best fit" algorithm.

Hyperlinks

In PQA and clipping pages, referencing HTML pages and graphics is very similar to normal Web pages. You can link to local files, other PQAs, and other Palm VII applications. You can also link to files on the Internet.

Marking a named location within a PQA or clipping page (bookmark)

Use this syntax to name a target of an internal cross-reference (link and target on the same PQA or clipping page):

```
<a name="targetname">textstring</a>
```

Linking to local pages and local graphics from a PQA

Use this syntax to link from the top-level index page to another page in your PQA:

```
<a href="page.html">textstring</a>
```

The link to another page in the PQA can include subdirectories:

```
<a href="subdirectory/page.html">textstring</a>
```

Use this syntax to link to a graphic in your PQA:

```
<img src="image.gif">
```

or

```
<img src="subdirectory/image.gif">
```

Linking to a PQA, local pages, and local graphics from a clipping

Use this syntax to link to another PQA from a clipping page:

```
<a href="file:someapp.pqa">textstring</a>
```

Use the LocalIcon META tag in the PQA index page to store graphics and pages in the PQA for use by clippings. Use this

method of storage only if the pages and images are not referenced from anywhere in the PQA:

```
<meta name="localicon" content="image.gif">
```

```
<meta name="localicon" content="page.html">
```

Once you have used LocalIcon to store graphics and pages in the PQA, you can call them from a clipping page. As noted in "PQA Directory Structure" on page 27, your links from clipping pages to PQA pages cannot include subdirectories:

```
<a href="file:mypqa.pqa/page.html">textstring</a>
```

```
<img src="file:mypqa.pqa/image.gif">
```

Linking to remote pages and graphics over the air

Use this syntax to link to a remote page, either from a PQA or a clipping page:

```
<a href="http://www.companyname.com/
page.html">textstring<a/>
```

Use this syntax to link to a remote graphics file if you want to leave it up to the user to download the image. Using such links is not recommended since they are inherently costly to the user:

```
<a href="http://www.companyname.com/
image.gif">textstring<a/>
```

An over-the-air icon is automatically added to an HTTP link:



Use this syntax for an HTTPS link:

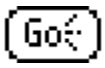
```
<a href="https://www.companyname.com/
page.html">textstring<a/>
```

A secure over-the-air icon is automatically added to an HTTPS link:



Linking from a BUTTON

You can use a special BUTTON attribute in PQA and clipping pages to make a button with a label and a hyperlink. A button can link to local or remote pages or graphics:



Use this syntax to link via a button to a local page in your PQA:

```
<a href="page.html" button>label</a>
```

Use this syntax to link via a button to a remote page:

```
<a href="http://www.companyname.com/page.html"
button>label</a>
```

Linking to a Palm OS application with PALM and PALMCALL

Use this syntax to launch another Palm OS application from a PQA or clipping page; *creatorid* is the four-digit Palm creator ID, not the application name:

```
<a href="palm:creatorid.appl">textstring</a>
```

Use this syntax to sublaunch another Palm OS application:

```
<a href="palmcall:creatorid.appl">textstring</a>
```

You can also pass arguments to the application you're calling; for example:

```
<a href="palmcall:memo.appl?name=john doe
address=office bldg"></a>
```

See also *Palm OS Programmer's Companion.*

Linking to the iMessenger application

Use this syntax to call the iMessenger application:

```
<a href="mailto:emailaddress">textstring</a>
```

You can also pass arguments, as in this example:

```
<a
href="mailto:johndoe@fixit.org?subject=lunch?body=
Give a call, John">Mail This</a>
```

See also *Palm OS Programmer's Companion.*

Query Form Elements

This section focuses on the usage in PQA pages of certain common form (<FORM>) elements, such as text fields, buttons, menus, and tables. No attempt has been made to present all the HTML tags and form elements supported in PQAs. See "HTML and Palm OS Extensions for PQAs" for more information on HTML tags.

As noted in "Minimizing Size of PQAs, Clippings, and Queries" on page 26, you can you can reduce the size of queries sent up to the Web Clipping proxy server by keeping variables in form elements brief and in lowercase. Similarly, HTML markup should be used only where necessary. Here, as elsewhere, every byte counts.

Text Fields and Text Areas

A text field (<INPUT TYPE="TEXT">) displays one line of editable text. You should specify the width of the text field with the SIZE attribute.

You can use text fields to input a name, an address, a ticker symbol, a quantity, a ZIP code, etc.

For best results, on the query form right-align the label and left-align the editable field. See .

For longer text, use a text area (<TEXTAREA>), instead. Text areas can scroll down as the user writes.

Pop-up Menus, Radio Buttons, and Check Boxes

Use pop-up menus (<SELECT>) to enter data such a state, city, airline, cuisine, price ranges, and other known categories.

While there are no technical limits on the number of options you can include on a menu, you should keep the user experience in mind. Users do not like scrolling a lot to find an option. And the more items on the menu, the slower the application runs. Around 50 items is probably the upper practical limit.

On the other hand, if you are providing only a few items, radio buttons (<INPUT TYPE="RADIO">) or check boxes (<INPUT TYPE="CHECKBOX">) may be better solutions. Use radio buttons for exclusive choices and check boxes for nonexclusive choices.

Keep radio button and check box labels short enough to look good on the Palm VII screen. Note that labels are rendered inside buttons on pages that use the PalmComputingPlatform META tag (as all your pages should). For more on the PalmComputingPlatform tag, see "Basic Structural Elements" on page 28.

When creating a pop-up menu of locations, make "current location" the default to take advantage of the Palm VII organizer's ability to send the handheld's current ZIP code to the Palm proxy server. See "Using Palm-specific Variables in Forms" on page 40.

Users can select menu options by entering Graffiti® characters. To make this feature work correctly, sort the menu options alphabetically. Do not put spaces or special characters before the option label.

Submit Button

Use submit buttons (<INPUT TYPE="SUBMIT">) to send information collected in the query form.

When naming submit buttons, make sure the label doesn't grow too large. Use terms that clearly indicate the action that will be taken; for example, "Find Package," "Get Directions," "Locate Office."

Using Palm-specific Variables in Forms

DATEPICKER, TIMEPICKER, %ZIPCODE, and %DEVICEID can supply useful functionality to your PQA queries. See Chapter 4, "Security and Authentication," for special uses of %DEVICEID.

DATEPICKER

Syntax:

```
<input type="datepicker" name="date"
value="value">
```

Use DATEPICKER to supply the current date or assign a value to the date. DATEPICKER returns the date in YYYY-MM-DD form (but the date is displayed to the user in MM/DD/YY form, unless overridden by user preferences). The current date is returned if VALUE is left undefined.

TIMEPICKER

Syntax:

```
<input type="timepicker" name="time"
value="value">
```

Use TIMEPICKER to supply the current time or assign a value to the time. TIMEPICKER returns the time in HH:MM 24-hour-clock format, but displays the time in HH:MM:AM/PM form on the Palm VII screen, unless overridden by user preferences. The current time is returned if VALUE is left undefined.

%ZIPCODE

Syntax:

```
<input type="hidden" value="%zipcode" name="zip">
```

%ZIPCODE is filled in when a form is submitted or a link is followed from the organizer. %ZIPCODE is replaced with the ZIP code of the nearest base station (usually within five to ten miles of the user).

%DEVICEID

Syntax:

```
<input type="hidden" value="%deviceid" name="id">
```

%DEVICEID is used to determine whether a query is coming from a valid Palm VII organizer. You can also use it to match a query with particular organizers and users.

These are the possible %DEVICEID return values:

- 1 – Proxy server recognizes device ID as valid
- 0 – May or may not be a Palm VII organizer
- -1 – Is not a Palm VII organizer
- x – Identifying string unique to user and organizer (set at device activation)

%ZIPCODE and %DEVICEID in URL

Syntax:

```
<form method="get" action="http://www.company.com/
cgi?zip=%zipcode&id=%deviceid">
```

You can also directly include %ZIPCODE and %DEVICEID at the end of any URL string.

Graphics

Use graphics when they effectively communicate a message or make things simpler for the user. Avoid using unnecessary graphics, especially over the air. Users will not want to pay for their transmission, and graphics slow down response time.

Graphics should be clear, bold, and easy to understand. Because of limitations of the handheld computer's screen, graphics should be as simple as possible.

File and Image Sizes; Color

To display in a PQA or clipping page, graphics can be no larger than a full page: 63 KB, 153 pixels wide, with a depth of 1 or 2 bits. This is a technical limit set by the system and screen size. Your graphics files should, of course, be as small as possible. The supported gray-scale colors are white, black, silver, and gray.

Icons for display in the Applications Launcher list view should be 15x9 pixels. Icons for display in the Applications Launcher icon view should be 32x22 pixels. Every PQA should contain these two images. If you don't specify your own icons, the Query Application Builder supplies default icons. See Chapter 5, "Building and Testing PQAs," for information on using the Query Application Builder.

File Types

The Palm VII organizer supports JPEG and GIF files for PQA and clipping pages. You can also include BMP files for large and small icons in the Applications Launcher (although these files can also be in GIF or JPEG format).

Sequential GIF animation is not supported.

Using the ALT Text Tag

If images are not displayable, the ALT text will be displayed, instead. If no ALT text is available, the word “image” will be displayed.

Stretching an Image

To “stretch” an image, you must create a separate image file for each size required. The Palm Image Checker can help you do that. See See Chapter 5, “Building and Testing PQAs,” for information on using the Image Checker.

Don’t try to use IMG tags with SIZE attributes different from those of the image to be displayed. This practice will not produce the results you would expect in a desktop Web browser.

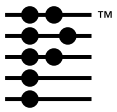
Linking Graphics

Graphics can be stored locally or sent over the air with a clipping. It is much better to store images locally whenever possible, for local pages and for clippings.

If you want to store an image file that is not referenced from any local page in a PQA for later reference by a clipping page, use the LocalIcon META tag in the PQA index page. This ensures that the image will be built into the PQA file and will be accessible from the clipping page.

See “Hyperlinks” on page 36 for link syntax and use of the LocalIcon META tag.

4



Security and Authentication

Security

The Palm VII organizer is designed to provide end-to-end secure information access. For public-key cryptographic functions, the elliptic-curve cryptographic system pioneered by Certicom, Inc., was selected by Palm Computing because of its security, efficient software implementation, operating speed, and small message size. Because of its embedded security mechanisms, the Palm VII organizer is a suitable platform for high-value applications such as electronic commerce. Not that while the secure message size using this technology is smaller than a message sent using other technologies, secure transactions inherently increase the size of a data packet. Therefore, the transmission of a secure message over the network will be slower than the transmission of an unsecured message.

The Palm VII organizer provides true end-to-end message security. When a sensitive message is sent to the network, it is encrypted and protected by a message integrity check (MIC). The MIC is used to detect transmission errors or manipulation, and the encryption prevents eavesdropping.

For more information on elliptic-curve cryptography, see Certicom, Inc.'s Web site at http://www.certicom.com.

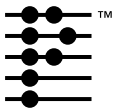
Authentication

Server authentication prevents the wireless session between the Palm VII organizer and the Web Clipping proxy server from being hijacked or spoofed. Communication between the Web Clipping proxy server and Internet servers is secured via Secure Sockets Layer (SSL) encryption and authentication.

Client authentication should be provided by your application. You can achieve this by requiring the user to enter a user ID and password when submitting a query. Client authentication can also be accomplished by including the %DEVICEID string in a hidden form field. See "Using Palm-specific Variables in Forms" on page 40.

If you are concerned with client authentication, we strongly encourage you to implement a time-out feature tied to the client's %DEVICEID string. For example, you could decide that a server will maintain an open connection for ten minutes after the last action the user performs. During that period, the user can go back through the application without having to reenter a password. After the ten-minute period expires, client authentication is required.

5



Building and Testing PQAs

This chapter provides instructions on using the Query Application Builder to create PQAs from HTML files and graphics. It is primarily procedural.

Make sure you read other parts of this book. Chapter 2, “Design Guidelines.” and Chapter 3, “PQA and Clipping Interface Elements.” contain very important information. Building a PQA is easy—but conceptualizing and planning are necessary to create a good PQA.

The Query Application Builder can be downloaded from http://www.palm.com/devzone/palmvii/. It runs under Windows 95, Windows 98, Windows NT, and Mac OS (System 7.x–8.x). Included in the downloaded Windows archive is the Palm Image Checker. This application is built in to the Mac OS version of the Query Application Builder.

Before you use the Query Application Builder, you need to

- Create HTML pages that conform to the PQA style guidelines presented in this book. One of these pages should be constructed as a top-level “index” page (though its name is not important).
- Make sure all pages and graphics have unique names. Don’t rely on file extensions to differentiate between names; for example, the Query Application Builder will see `Myapp.html` and `Myapp.gif` as the same name. Your HTML files can compile with either `.htm` or `.html` extensions.
- Put all the components that will make up the PQA under one root directory. Subdirectories are permitted for purposes of building a PQA, but not for purposes of linking between pages and graphics within the PQA from clipping pages.

- Prepare graphics for use in the PQA. See "Graphics" on page 42. Preview your images in the Image Checker to see how they will look on the Palm organizer's screen.

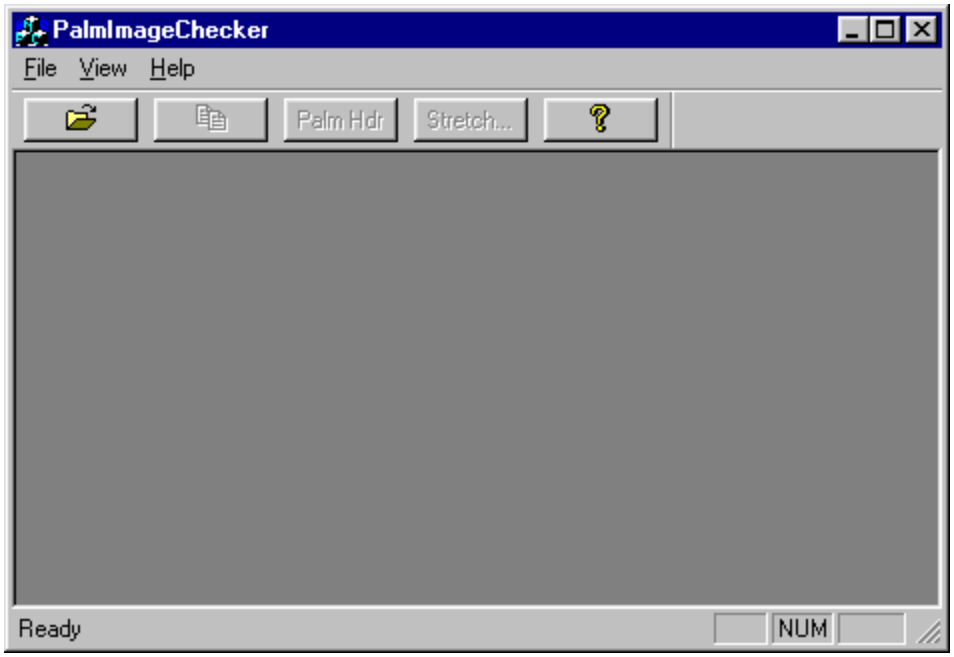
Using the Image Checker

You use the Image Checker to preview GIF, JPEG, and BMP files to see how they will look on the Palm VII organizer.

To Preview an Image

1. **Start the Image Checker**

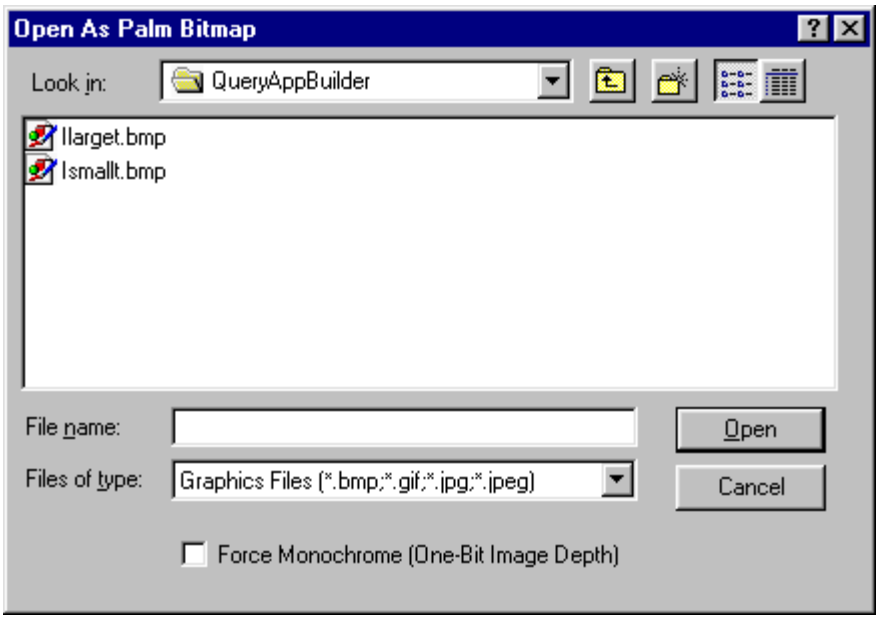
   Double-click the icon or choose Start>Run and find `pic.exe` in the directory where you installed the Image Checker.



2. **Click Open.**

   You can also press Ctrl+O or choose File>Open.

   The Open As Palm Bitmap dialog box is displayed.



3. **[Optional] Check Force Monochrome (One-Bit Image Depth).**

   This displays in the graphics file in black and white instead of the default Palm gray scale.

4. **Choose a file to preview and click Open.**

   The Image Checker converts the file to Palm gray scale—or black and white, if you checked Force Monochrome, above—and displays the image as it would appear on the Palm organizer.

5. **[Optional] To review the file description, click Palm Hdr.**

   You can also choose Palm Bitmap>Show Palm Header.

6. **To quit the Image Checker, choose File>Exit.**

   If necessary, use graphics-editing software to alter the image and then preview it again in the Image Checker. You can only preview, copy, and resize graphics files in the Image Checker.

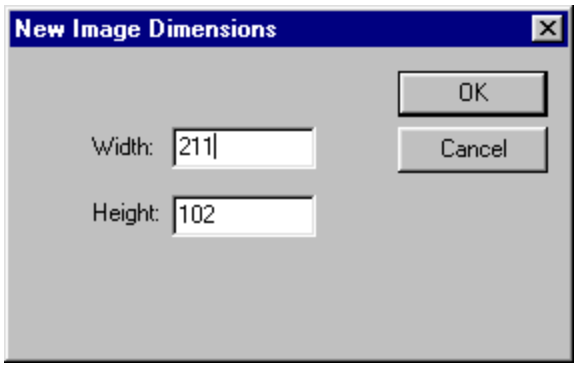
To Resize an Image in the Image Checker

If you need to resize an image, Image Checker can help. You can resize the image, copy it to the clipboard, and then edit or save it in a graphics-editing program.

Measurements are in pixels. Make sure they conform to style guidelines.

1. **Start the Image Checker and open a file to preview.**
2. **Click Stretch.**

   The New Image Dimensions dialog box shows the current width and height.



3. **Enter new height or width measurements.**

   These measurements should conform to style guidelines. See "Graphics" on page 42.

   To maintain proportion, enter 0 (zero) as either the new height or the new width. The Image Checker will recalculate that zero value based on the figure you enter for the other measurement.

4. **Click OK.**

   The Image Checker displays the resized graphic in a new window.

5. **Click Copy.**

   You can also press Ctrl+C or choose Edit>Copy. This copies the resized image to the Windows clipboard.

6. **Paste the resized image into a graphics program to save it or edit it.**
7. **Preview the edited image again in the Image Checker.**

Using the Query Application Builder

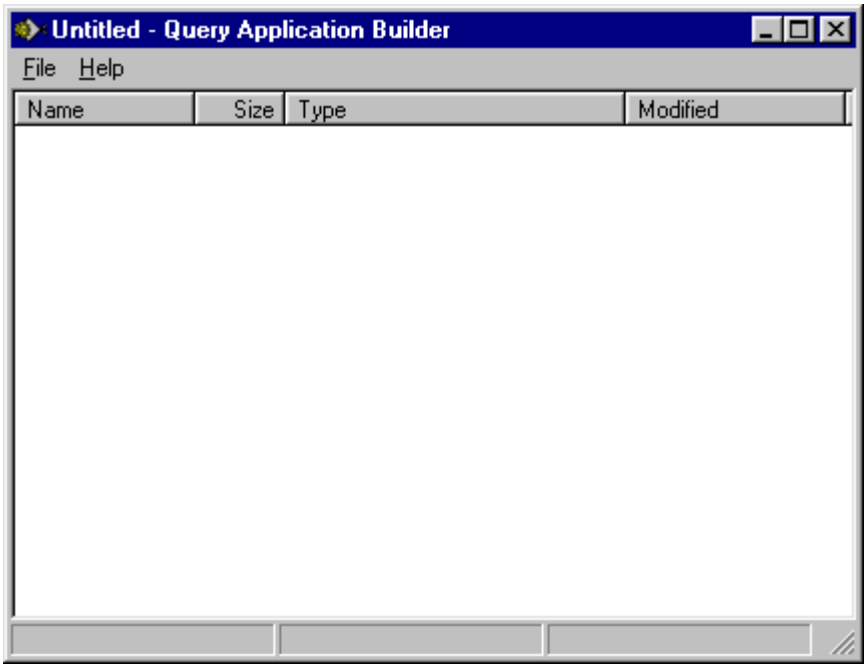
Once you have gathered all the PQA components under a root directory, you use the Query Application Builder to create the .pqa file, a user-installable Palm database.

The steps shown here are for the Windows version of the Query Application Builder, but the Mac OS version has similar functionality.

To Build a PQA

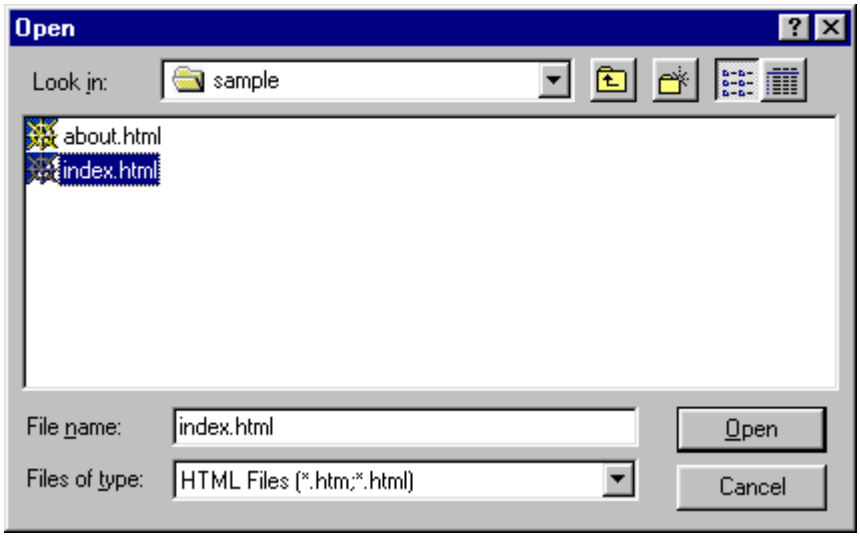
1. **Start the Query Application Builder.**

   Double-click the icon or choose Start>Run and find `qab.exe` in the directory where you installed the Query Application Builder.



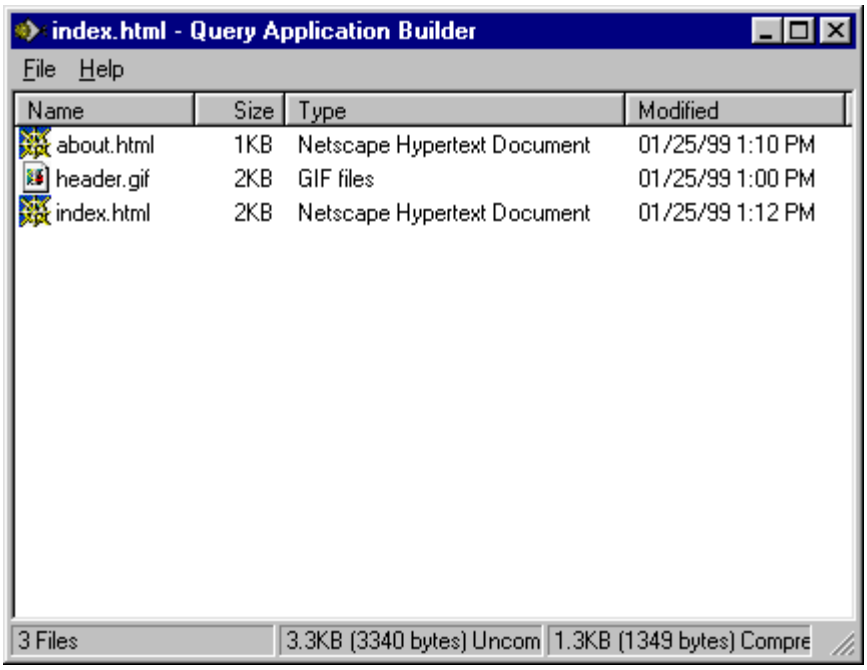
2. **Choose File>Open Index.**

   The Open dialog box is displayed.



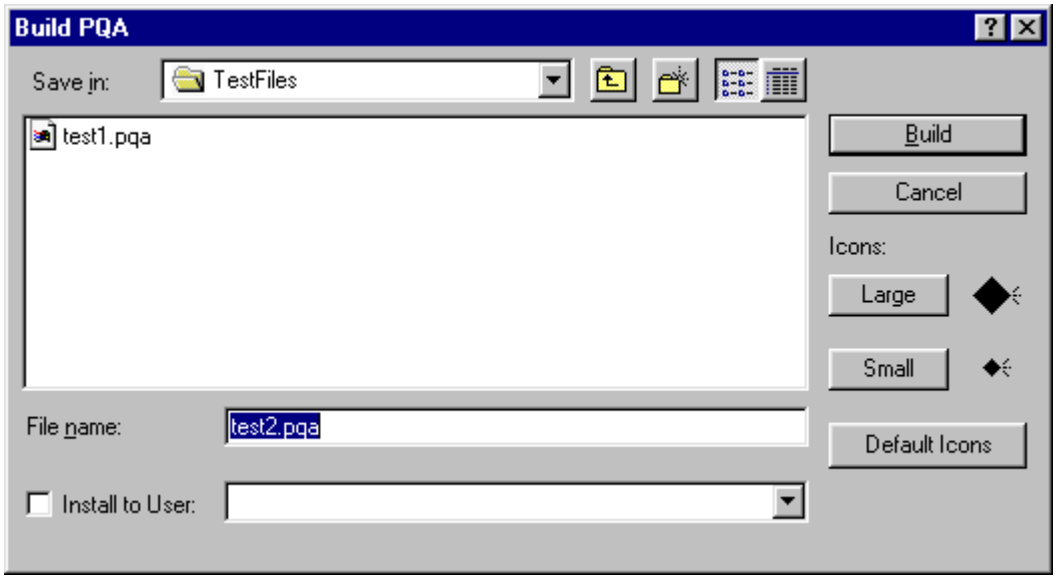
3. **Choose the HTML page that is to be the top-level page of the PQA and click Open.**

   All the components of the PQA are listed. If a file is not listed here, it will not be included in the PQA build. If you think a file is missing from the list, check the links in the HTML files and make sure the correct Palm META tags have been used. See "Basic Structural Elements" on page 28 and "Hyperlinks" on page 36. Note that the sizes of the uncompressed and compressed PQA components are displayed in the Status bar.



4. Choose File>Build PQA.

The Build PQA dialog box is displayed.



5. Specify a location to save the PQA file.

6. **Type the name of the PQA in the File name field.**

   This is the name that identifies the PQA on the Palm VII organizer. PQAs take the .pqa extension, but this extension is not displayed in the Applications Launcher. In the Mac OS version of the Query Application Builder, options enable you to assign a different name to the file and to the application as it appears in the Application Launcher. In the Windows version of the Query Application Builder, you can change the name with a command-line argument. See "Naming Your PQA" on page 55 and "Query Application Builder Command-line Interface" on page 55.

7. **Click Icons: Large/Small to specify the large and small icons.**

   These are the icons that identify the PQA in the Applications Launcher icon view and list view. If you don't supply icons, the Query Application Builder supplies default icons.

   a. **In the Build PQA dialog box, click Large.**

   b. **Specify the icon to be displayed in the Applications Launcher icon view and click Open.**

      The image is displayed in the Build PQA dialog box, next to Large

   c. **In the Build PQA dialog box, click Small.**

   d. **Specify the icon to be displayed in the Applications Launcher list view and click Open.**

      The image is displayed in the Build PQA dialog box, next to Small.

8. **[Optional] To place the PQA in the Install Folder for the user shown, check Install to User and choose a user name.**

   The next HotSync operation installs the application on the user's Palm VII organizer.

9. **Click Build.**

   The Query Application Builder compiles all the local hyperlinks in the HTML file you designated as the top-level page. It creates a .pqa file with the name you specified and stores in it the location you specified. It (optionally) adds the .pqa file to the Install Folder for the designated user.

Naming Your PQA

When you name your PQA in the Build PQA window of the Query Application Builder, that name is displayed when the PQA is installed from the desktop and in the Applications Launcher on the Palm VII organizer. The name displayed in the Applications Launcher does not have the `.pqa` extension, though. So a file named `FineFix.pqa` in the Build PQA window appears as `FineFix.pqa` in the installer and FineFix in the Applications Launcher.

However, you can give the file a second name, which will appear only in the Applications Launcher. And you can even give the file a third name, which is used in the installer or when downloading from the Internet. So a file named `FineFix.pqa` in the Build PQA window could appear as `CurrentFineFix` in the installer and Fine Fix in the Applications Launcher. The installer name and the Applications Launcher name can use special characters, something not allowed in the "basic" `.pqa` file name.

The graphical user interface of the Mac OS version of the Query Application Builder enables you manipulate the PQA names, but the Windows version does not. Instead, you can use the command-line interface described in the next section.

Query Application Builder Command-line Interface

You can also run the Query Application Builder from the Windows command line. (The Mac OS equivalents for some of these commands are built in to the Query Application Builder graphical user interface.)

1. **Choose Start>Run.**
2. **Type qab followed by the appropriate parameters.**

   Use double quotation marks around file names, as shown.

Parameters for qab	Meaning
`/h`	Displays Help dialog box
`/pqa "`*`index`*`.html"`	Builds a PQA
`/n "`*`pqaname`*`"`	Specifies the PQA name to be displayed in Applications Launcher; this makes the PQA name different from the `.pqa` file name. Names longer than eight characters may be truncated. You can include special characters like * and @.
`/o "`*`output`*`.pqa"`	Specifies the output file name
`/l "`*`largeicon`*`.gif"`	Specifies the graphic to use as large icon; use BMP, GIF, or JPEG file
`/s "`*`smallicon`*`.gif"`	Specifies the graphic to use as small icon; use BMP, GIF, or JPEG file
`/u "`*`username`*`"`	Specifies user folder into which to install PQA
`/v`	Displays error strings (verbose mode)

Testing the PQA on POSE and Palm VII Organizer

You can test the functionality of your PQA either by installing it on the Palm OS Emulator (with the Palm VII ROM) or by installing it on a Palm VII organizer. Once you have installed your PQA, you can test links to local pages and graphics by tapping or clicking.

To test the PQA-clippings functionality, you need to send queries or follow hyperlinks over the air. You can do this with a Palm VII

organizer simply by launching the PQA and submitting the query form or tapping remote hyperlinks. Queries will go out over the air on your Palm.Net account.

You can also test over-the-air links with the Palm OS Emulator running on a desktop computer with a connection to the Internet. (You can download POSE from http://www.palm.com/devzone/pose/.) This connection is free of charge for testing purposes. See the instructions below.

To Install the PQA on the Palm OS Emulator

These steps are for the Windows version of the software. Mac OS functionality is similar.

1. **Start POSE.**
2. **Drag the PQA file and drop it on the Applications Launcher window.**

   The PQA is installed for the current session. Now you can test its appearance and behavior.

To Test the PQA Over the Internet with POSE

If your desktop computer has a connection to the Internet and you have downloaded and installed the Palm OS Emulator, you can use Palm's special IP address test your PQAs and clippings.

These steps are for the Windows version of the software. Mac OS functionality is similar.

1. **Start POSE.**
2. **Verify that Wireless Preferences are set correctly.**

   You can test using the Palm proxy server free of charge: at the time of writing the IP address is 207.240.80.136, but that address is subject to change. Verify the current IP address at http://www.palm.com/devzone/palmvii/.

3. **If you are dialing up through modem, open the POSE Properties window to set the Serial Port and check Redirect NetLib calls to host TCP/IP.**

To Install the PQA on a Palm VII Organizer

These steps are for the Windows version of the software. Mac OS functionality is similar.

1. **Start Palm™ Desktop software.**
2. **Click Install.**

   The Install Tool is displayed.
3. **Click Add.**

   The Open dialog box is displayed.
4. **Specify a PQA file and click Open.**

   The PQA is appears in the Install Folder.
5. **Click Done.**
6. **Place your Palm VII organizer in the cradle and press the HotSync button.**

   Now you can test the appearance and behavior of your PQA.

A

HTML and Palm OS Extensions for PQAs

This appendix provides tables of all HTML markup compliant with the Web Clipping application of the Palm VII organizer. It also includes Palm OS extensions. The use of many of these elements, attributes, and strings is discussed in Chapter 3, "PQA and Clipping Interface Elements." See "Palm OS HTML 3.2 DTD" for information on character entities.

Valid markup for PQA and clippings pages is shown in the following tables.

Table A.1 Structural Tags

Tag	Description	Attributes	Usage Notes
<HTML>	Indicates beginning of HTML source		*Highly recommended in all documents*
<HEAD>	Encloses the document header		*Highly recommended in all documents*
<TITLE>	Text appears in Navigation bar		*Highly recommended in all documents*
<META>	Used to include document-level information	HTTP-EQUIV, NAME, CONTENT	See Table A.2 below for Palm-specific META tags.

Table A.1 Structural Tags

Tag	Description	Attributes	Usage Notes
<BODY>	Document content	BGCOLOR, TEXT	*Required in all documents.* BGCOLOR and TEXT optional to specify colors. Four colors are available: BLACK (#000000), SILVER (#C0C0C0), GRAY (#808080), WHITE (#FFFFFF)

Table A.2 Special Palm OS Elements, Attributes, and Strings

Tag	Description	Usage Notes
HistoryListText	META tag that populates History list	*Highly recommended for all clipping pages.* If <META NAME="HistoryListText" CONTENT="My Stocks &date &time">, then “My Stocks” is displayed in History list along with date and time.
LocalIcon	META tag that includes local graphics and HTML pages in PQA	*Use only in PQA pages.* Images and pages stored with LocalIcon can be linked to from clipping pages. Store an image in a PQA with <META NAME="LocalIcon" CONTENT="*image*.gif"> ; in clipping page, link to stored image with <IMG SRC="file:*mypqa*.pqa/*image*.gif">.

Table A.2 Special Palm OS Elements, Attributes, and Strings

Tag	Description	Usage Notes
PalmComputingPlatform	META tag that identifies page as Palm Friendly	*Required in all documents.* Use in all pages to ensure proper display of text and images: <META NAME="PalmComputingPlatform" CONTENT="true">
PalmLauncherRevision	META tag that includes version number in PQA	*Highly recommended for index page of PQA.* Very useful when upgrading PQA. <META NAME="PalmLauncerRevision" CONTENT="*x.x*">, with *x.x* representing version number.
<SMALLSCREENIGNORE>	Marks areas that shouldn't be displayed on PalmVII organizer	*Use in clipping pages.* Content placed between start and end tags will be ignored by the Clipping proxy server.
BUTTON	<A> attribute that produces a a labeled hyperlink button to local or remote file	<A HREF="http:/ / *yrcompany.com* / moreinfo.html" BUTTON>Get Info<A /> produces a hyperlink button with label, "Get Info." See Chapter 3, "PQA and Clipping Interface Elements.".

Table A.2 Special Palm OS Elements, Attributes, and Strings

Tag	Description	Usage Notes
%ZIPCODE	String returns ZIP code of base station	When the Clipping proxy server encounters %ZIPCODE, it replaces the string with the 5-digit ZIP code of the base station. Some base stations overlap ZIP codes, so accuracy is not guaranteed. Requests coming from an emulated Palm VII organizer or from a base station whose ZIP code has not been entered, return a value of 00000.
%DEVICEID	String returns unique Palm VII organizer ID	When the client encounters %DEVICEID, it replaces the string with the Palm VII organizer's ID. Return values indicate a Palm device (1), not a Palm device (-1), or undetermined device (0).
TIMEPICKER	FORM INPUT TYPE takes and displays time	In <INPUT TYPE=TIMEPICKER NAME="Time1" VALUE="20:00">, TIMEPICKER takes the time in HH:MM or HH:MM:AM/PM format; display can be set by user preferences to include AM and PM. If no VALUE given, current time is supplied.

Table A.2 Special Palm OS Elements, Attributes, and Strings

Tag	Description	Usage Notes
DATEPICKER	FORM INPUT TYPE takes and displays date	In <INPUT TYPE=DATEPICKER NAME="Date1" VALUE="06/15/99">, DATEPICKER takes the date in MM/DD/YY format and sends it over the air in YYYY-MM-DD format. If no VALUE given, then current date is supplied.

Table A.3 Block Tags

Tag	Description	Attributes	Usage Notes
<ADDRESS>	Encloses author and contact details		
<BLOCKQUOTE>	Indents block quotations		
 	Forces line break	CLEAR	CLEAR moves break down past floating images on either or both margins.
<CENTER>	Same as <DIV ALIGN=CENTER>		
<DIV>	Structures document with hierarchical divisions; breaks up <P> with a new line	ALIGN	
<H1>–<H6>	Hierarchical documents headings	ALIGN	See font tables in Chapter 3, "PQA and Clipping Interface Elements."

Table A.3 Block Tags

Tag	Description	Attributes	Usage Notes
<HR>	Horizontal rule	ALIGN, SIZE, WIDTH	SIZE changes thickness of line. WIDTH defaults to 50%. Values can be in number of pixels or percentage.
<LISTING>	Includes preformatted ASCII text	None	Deprecated but supported. See PRE.
<P>	Paragraph break	ALIGN	
<PLAINTEXT>	Includes preformatted ASCII text	None	Deprecated but supported. See PRE.
<PRE>	Includes preformatted ASCII text.	None	
<XMP>	Includes preformatted ASCII text	None	Deprecated but supported. See PRE.

Table A.4 Formatting Tags

Tag	Description	Usage Notes
<B>	Boldface	For effects of formatting tags on fonts, see Chapter 3, "PQA and Clipping Interface Elements.".
<BIG>	Increase size	
<CITE>	Italic	
<CODE>	Monospace	
<DFN>	Italic	
<EM>	Italic	
<FONT>	Varies	Can take SIZE and COLOR attributes. See "Fonts" on page 31.
<I>	Italic	
<KBD>	Monospace	
<S>	Strikethrough	
<SAMP>	Monospace	
<SMALL>	Decrease size	
<STRIKE>	Strikethrough	
<STRONG>	Boldface	
<TT>	Monospace	
<U>	Underline	
<VAR>	Italic	

Table A.5 Hypertext and Image Tags

Tag	Description	Attributes	Usage Notes
<A>	Hyperlink or anchor	NAME, HREF, TITLE, BUTTON	See Chapter 3, "PQA and Clipping Interface Elements." See also Table A.2 .
<IMG>	Inserts image	SRC, ALT, HEIGHT, WIDTH, ALIGN, BORDER, HSPACE, VSPACE	See Chapter 3, "PQA and Clipping Interface Elements."
<BASE>	BASE URL dereferences relative URLs	HREF=*url*	Use to dereference relative URLs. Of limited utility in PQAs and clippings.

Table A.6 List Tags

Tag	Description	Attributes	Usage Notes
<UL>	Unordered list	TYPE, START	Maximum five levels are displayed; nested lists are supported to width of screen; use with <LI>
<OL>	Ordered list	TYPE, START	Maximum five levels are displayed; nested lists are supported to width of screen; use with <LI>
<LI>	List item	TYPE, VALUE	Use with <UL> or <OL>

Table A.6 List Tags

Tag	Description	Attributes	Usage Notes
<DL>	Definition list		Use with <DT> and <DD>
<DT>	Definition title		Use with <DL> and <DD>
<DD>	Definition description		Use with <DL> and <DT>

Table A.7 Table Tags

Tag	Description	Attributes	Usage Notes
<TABLE>	Starts and ends table	ALIGN, WIDTH, BORDER, CELLPADDING, CELLSPACING	See Chapter 3, "PQA and Clipping Interface Elements." Tables are rendered without a border, unless BORDER is specified. Tables are sized to "best-fit" unless WIDTH is set in pixels. But tables wider than 153 pixels will be truncated at column divisions. Nested tables are not supported: only outermost table is rendered as such.
<TR>	Table row	ALIGN, VALIGN	
<TD>	Table cell (data)	ALIGN, ROWSPAN, WIDTH, COLSPAN, HEIGHT	

Table A.7 Table Tags

Tag	Description	Attributes	Usage Notes
<TH>	Table cell (header)	ALIGN, ROWSPAN, WIDTH, COLSPAN, HEIGHT	
<CAPTION>	Table caption	ALIGN	

Table A.8 Form Tags

Tag	Description	Attributes	Usage Notes
<FORM>	Begins and ends form	METHOD, ACTION, ENCTYPE	POST and GET methods are supported as is the default ENCTYPE of application/x-wwww-form-url-encoded.
<INPUT>	Specifies input fields in form	TYPE, NAME, VALUE, CHECKED, SIZE, MAXLENGTH, SRC, ALIGN	TYPE may be TEXT, PASSWORD, CHECKBOX, RADIO, SUBMIT, RESET, HIDDEN, TIMEPICKER, DATEPICKER. For TYPE attributes, see Table A.9 . See also Chapter 3, "PQA and Clipping Interface Elements." See also Table A.2 .
<TEXTAREA>	Multiline input field	ROWS, COLS, NAME	
<SELECT>	Multiple-choice selections in pop-up menu	NAME, SIZE, MULTIPLE	
<OPTION>	Item in pop-up menu	SELECTED, VALUE	

Table A.9 Form Tags – <INPUT TYPE> Attributes

INPUT TYPE	Description	Attributes	Usage Notes
TEXT	Single-line text field	NAME, VALUE, SIZE, MAXLENGTH	All these tags are to be used in forms with the syntax <INPUT TYPE=TEXT SIZE=8>.
PASSWORD	Single-line text field that echoes input characters with *	NAME, VALUE, SIZE, MAXLENGTH	
CHECKBOX	Check boxes for multiple selections	NAME, VALUE, CHECKED	
RADIO	Radio buttons for exclusive selection	NAME, VALUE, CHECKED	
SUBMIT	Submit button to send form contents to server	NAME, VALUE	
HIDDEN	Unrendered field whose information is to be sent back to the server on submission of form	NAME, VALUE	
RESET	Resets form to initial state	NAME, VALUE	

B

Palm OS HTML 3.2 DTD

This appendix provides the complete Palm OS HTML 3.2 Document Type Description. Much of this information is covered in Chapter 3, "PQA and Clipping Interface Elements," and "HTML and Palm OS Extensions for PQAs" in more accessible fashion.

Listing B.1 Palm OS HTML 3.2 DTD

```
Palm OS HTML 3.2 Document Type Definition
<!--
        Palm OS Document Type Definition for the HyperText Markup
        Language version 3.2 as implemented for the Web Clipping
        application of Palm OS 3.2 and the Clipping Proxy Server.

        For more information on Palm OS look at URL http://
www.palm.com

        Date: Monday June 7th 1999

       Author: Scot Stennis <scot_stennis@palm.com> based on the
                W3C HTML 3.2 DTD by Dave Raggett <dsr@w3.org>

        Palm OS HTML 3.2 aims to capture supported W3C HTML 3.2
elements
        and attributes as well as Palm OS extensions.-->

<!ENTITY % HTML.Version
        "-//POS//DTD Palm OS HTML 3.2 Final//EN"

        -- Typical usage (optional):
```

```
            <!DOCTYPE HTML PUBLIC "-//POS//DTD Palm OS HTML 3.2
Final//EN">
            <html>
            ...
            </html>
        --
        >


<!--================== Deprecated Features Switch
==========================-->

<!ENTITY % HTML.Deprecated "INCLUDE">


<!--================== Imported Names
====================================-->

<!ENTITY % Content-Type "CDATA"
        -- meaning a MIME content type, as per RFC1521
        -->

<!ENTITY % HTTP-Method "GET | POST"
        -- as per HTTP specification
        -->

<!ENTITY % URL "CDATA"
        -- The term URL means a CDATA attribute
           whose value is a Uniform Resource Locator,
           See RFC1808 (June 95) and RFC1738 (Dec 94).
        -->

<!-- Parameter Entities -->

<!ENTITY % head.misc "META" -- repeatable head elements -->

<!ENTITY % heading "H1|H2|H3|H4|H5|H6">

<!-- DIR and MENU behave as UL -->
<!ENTITY % list "UL | OL | DIR | MENU">
```

```
<![ %HTML.Deprecated [
    <!ENTITY % preformatted "PRE | XMP | LISTING">
]]>

<!ENTITY % preformatted "PRE">


<!--================ Character mnemonic entities
=========================-->

<!ENTITY % ISOlat1 PUBLIC
       "ISO 8879-1986//ENTITIES Added Latin 1//EN//HTML">
%ISOlat1;


<!--================ Entities for special symbols
=========================-->
<!-- Some common symbols ( e.g.: &reg; ): -->

<!ENTITY amp    CDATA "&#38;"   -- ampersand             -->
<!ENTITY gt     CDATA "&#62;"   -- greater than          -->
<!ENTITY lt     CDATA "&#60;"   -- less than             -->
<!ENTITY copy   CDATA "&#169;"  -- copyright             -->
<!ENTITY reg    CDATA "&#174;"  -- registered trademark  -->


<!--=================== Text Markup
=====================================-->

<!-- S and STRIKE are synonymous -->
<!ENTITY % font "TT | I | B  | U | STRIKE | S | BIG | SMALL">

<!-- EM, DFN, VAR, CITE have similar behavior;
     CODE, SAMP, KBD have similar behavior;
     STRONG behaves as B
-->
<!ENTITY % phrase "EM | STRONG | DFN | CODE | SAMP | KBD | VAR |
CITE">
```

```
<!ENTITY % special "A | IMG | FONT | BR">

<!ENTITY % form "INPUT | SELECT | TEXTAREA">

<!ENTITY % text "#PCDATA | %font | %phrase | %special | %form">

<!ELEMENT (%font|%phrase) - - (%text)*>

<!-- there are also 4 supported color names:

   black, gray, silver, and white

 -->

<!ELEMENT FONT - - (%text)*     -- local change to font -->
<!ATTLIST FONT
    size    CDATA   #IMPLIED    -- [+]nn e.g. size="+1", size=4 -
-
    color   CDATA   #IMPLIED    -- #RRGGBB in hex, e.g. black:
color="#000000"
                                  4 colors are supported: black
(#000000),
                                                          silver
(#C0C0C0),
                                                          gray
(#808080),
                                                          white
(#FFFFFF) --
    >

<!ELEMENT BASEFONT - O EMPTY     -- base font size (1 to 6) -->
<!ATTLIST BASEFONT
        size  CDATA  #IMPLIED    -- e.g.: size=3 --
        >

<!ELEMENT BR    - O EMPTY    -- forced line break -->
<!ATTLIST BR
        clear (left|all|right|none) none -- control of text flow
--
        >
```

```
<!--================== HTML content models
================================-->
<!--
    HTML has three basic content models:

        %text           character level elements and text strings
        %flow           block-like elements e.g. paragraphs and
lists
        %body.content   as %flow plus headers H1-H6 and ADDRESS
-->

<!ENTITY % block
     "P | %list | %preformatted | DL | DIV | CENTER |
      BLOCKQUOTE | FORM | HR | TABLE">

<!-- %flow is used for DD and LI -->
<!ENTITY % flow "(%text | %block)*">


<!--================== Document Body
====================================-->

<!ENTITY % body.content "(%heading | %text | %block | ADDRESS)*">

<!ENTITY % color "CDATA" -- a color specification: #HHHHHH -->

<!ENTITY % body-color-attrs "
        bgcolor %color #IMPLIED
        text    %color #IMPLIED
        ">

<!ELEMENT BODY O O  %body.content>
<!ATTLIST BODY
        %body-color-attrs;  -- bgcolor, text --
        >

<!ENTITY % address.content "((%text;) | P)*">
```

```
<!ELEMENT ADDRESS - - %address.content>

<!ELEMENT DIV - - %body.content>
<!ATTLIST DIV
        align   (left|center|right) #IMPLIED -- alignment of
following text --
        >

<!-- CENTER is a shorthand for DIV with ALIGN=CENTER -->
<!ELEMENT center - - %body.content>


<!--================== The Anchor Element
================================-->

<!ELEMENT A - - (%text)* -(A)>
<!ATTLIST A
        name    CDATA   #IMPLIED    -- named link end --
        href    %URL    #IMPLIED    -- URL for linked resource --
        title   CDATA   #IMPLIED    -- advisory title string --
        >


<!--=================== Images
============================================-->

<!ENTITY % Length "CDATA"   -- nn for pixels or nn% for
percentage length -->
<!ENTITY % Pixels "NUMBER"  -- integer representing length in
pixels -->

<!-- Suggested widths are used for negotiating image size
     with the module responsible for painting the image.
     align=left or right cause image to float to margin
     and for subsequent text to wrap around image -->

<!ENTITY % IAlign "(top|middle|bottom|left|right)">

<!ELEMENT IMG    - O EMPTY --  Embedded image -->
<!ATTLIST IMG
```

```
        src     %URL     #REQUIRED  -- URL of image to embed --
        alt     CDATA    #IMPLIED   -- for display in place of
image --
        align   %IAlign  #IMPLIED   -- vertical or horizontal
alignment --
       height  %Pixels  #IMPLIED   -- suggested height in pixels
--
        width   %Pixels  #IMPLIED   -- suggested width in pixels
--
        >

<!--=================== Horizontal Rule
=================================-->

<!ELEMENT HR    - O EMPTY>
<!ATTLIST HR
        align (left|right|center) #IMPLIED
        size  %Pixels #IMPLIED
        width %Length #IMPLIED
        >

<!--===================
Paragraphs=============================================-->

<!ELEMENT P     - O (%text)*>
<!ATTLIST P
        align  (left|center|right) #IMPLIED
        >

<!--=================== Headings
=========================================-->

<!--
  There are six levels of headers from H1 (the most important)
  to H6 (the least important).
-->

<!ELEMENT ( %heading )  - -  (%text;)*>
<!ATTLIST ( %heading )
        align  (left|center|right) #IMPLIED
```

```
        >

<!--=================== Preformatted Text
================================-->

<!-- excludes images and changes in font size -->
<!ENTITY % pre.exclusion "IMG|BIG|SMALL|FONT">

<!ELEMENT PRE - - (%text)* -(%pre.exclusion)>

<![ %HTML.Deprecated [

<!ENTITY % literal "CDATA"
        -- historical, non-conforming parsing mode where
           the only markup signal is the end tag
           in full
        -->

<!ELEMENT (XMP|LISTING) - -  %literal>
<!ELEMENT PLAINTEXT - O %literal>

]]>


<!--=================== Block-like Quotes
================================-->

<!ELEMENT BLOCKQUOTE - - %body.content>


<!--=================== Lists
============================================-->

<!--
    Palm OS HTML 3.2 allows you to control the sequence number
for ordered lists.
    You can set the sequence number with the START and VALUE
attributes.
    The TYPE attribute may be used to specify the rendering of
ordered
```

```
    and unordered lists.
-->

<!-- definition lists - DT for term, DD for its definition -->
<!ELEMENT DL    - -  (DT|DD)+>
<!ATTLIST DL >

<!ELEMENT DT - O  (%text)*>
<!ELEMENT DD - O  %flow;>

<!-- Ordered lists OL, and unordered lists UL -->
<!ELEMENT (OL|UL) - -  (LI)+>

<!--
       Numbering style
    1   Arabic numbers      1, 2, 3, ...
    a   lower alpha         a, b, c, ...
    A   upper alpha         A, B, C, ...
    i   lower Roman         i, ii, iii, ...
    I   upper Roman         I, II, III, ...

    The style is applied to the sequence number which by default
    is reset to 1 for the first list item in an ordered list.

-->

<!ENTITY % OLStyle "CDATA" -- constrained to: [1|a|A|i|I] -->

<!ATTLIST OL -- ordered lists --
        type      %OLStyle   #IMPLIED   -- numbering style --
        start     NUMBER     #IMPLIED   -- starting sequence
number --
        >

<!-- bullet styles -->

<!-- disc and square are rendered as square; disc is the default
type-->
<!ENTITY % ULStyle "disc|square|circle">
```

```
<!ATTLIST UL -- unordered lists --
        type    (%ULStyle)   #IMPLIED   -- bullet style --
        >

<!ELEMENT (DIR|MENU) - -  (LI)+ -(%block)>
<!ATTLIST DIR
        >
<!ATTLIST MENU
        >

<!-- <DIR>              Directory list                 -->
<!-- <MENU>             Menu list                      -->

<!-- The type attribute can be used to change the bullet style
     in unordered lists and the numbering style in ordered lists
-->

<!ENTITY % LIStyle "CDATA" -- constrained to:
"(%ULStyle|%OLStyle)" -->

<!ELEMENT LI - O %flow -- list item -->
<!ATTLIST LI
        type    %LIStyle     #IMPLIED   -- list item style --
        value    NUMBER      #IMPLIED   -- reset sequence number
--
        >


<!--================ Forms
================================================-->

<!ELEMENT FORM - - %body.content -(FORM)>
<!ATTLIST FORM
        action  %URL #IMPLIED      -- server-side form handler --
        method  (%HTTP-Method) GET -- see HTTP specification --
        enctype %Content-Type; "application/x-www-form-
urlencoded"
        >

<!ENTITY % InputType
```

```
        "(TEXT | PASSWORD | CHECKBOX | RADIO | SUBMIT | RESET |
HIDDEN |
          DATEPICKER | TIMEPICKER)"
                                        -- DATEPICKER displays MM/DD/
YY format,
                                           but submits as YYYY-MM-DD
format.
                                           TIMEPICKER displays HH:MM
(am/pm)
                                         format, but submits 24 hour
clock
                                           HH:MM format. --
        >

<!ELEMENT INPUT - O EMPTY>
<!ATTLIST INPUT
        type      %InputType TEXT      -- what kind of widget is
needed --
        name      CDATA      #IMPLIED  -- required for all but
submit and reset --
        value     CDATA      #IMPLIED  -- required for radio and
checkboxes --
        checked   (checked)  #IMPLIED  -- for radio buttons and
check boxes --
        size      CDATA      #IMPLIED  -- specific to each type
of field --
        maxlength NUMBER     #IMPLIED  -- max chars allowed in
text fields --
        >

<!ELEMENT SELECT - - (OPTION+)>
<!ATTLIST SELECT
        name      CDATA     #REQUIRED
        size      NUMBER    #IMPLIED
        multiple (multiple) #IMPLIED -- size attribute should be
set to
                                        minimum of 2 for proper
display
                                        when using multiple --
        >
```

```
<!ELEMENT OPTION - O (#PCDATA)*>
<!ATTLIST OPTION
        selected (selected) #IMPLIED
        value  CDATA  #IMPLIED -- defaults to element content --
        >

<!-- Multi-line text input field. -->
<!ELEMENT TEXTAREA - - (#PCDATA)*>
<!ATTLIST TEXTAREA
        name CDATA #REQUIRED
        rows NUMBER #REQUIRED
        cols NUMBER #REQUIRED
        >


<!--======================= Tables
=======================================-->

<!-- Widely deployed subset of the full table standard, see RFC
1942
     e.g. at http://www.ics.uci.edu/pub/ietf/html/rfc1942.txt -->

<!-- horizontal placement of table relative to window -->
<!ENTITY % Where "(left|center|right)">

<!-- horizontal alignment attributes for cell contents -->
<!ENTITY % cell.halign
        "align  (left|center|right) #IMPLIED"
        >

<!ELEMENT table - - (caption?, tr+)>
<!ELEMENT tr - O (th|td)*>
<!ELEMENT (th|td) - O %body.content>

<!ATTLIST table                          -- table element --
        align       %Where;   #IMPLIED  -- table position
relative to window --
        width       %Length   #IMPLIED  -- table width relative
to window;
```

```
                                                percentage widths not
supporte --
        border      %Pixels   #IMPLIED  -- controls frame width
around table --
        cellspacing %Pixels   #IMPLIED  -- spacing between cells
--
        cellpadding %Pixels   #IMPLIED  -- spacing within cells -
-
        >

<!ELEMENT CAPTION - - (%text;)* -- table or figure caption -->
<!ATTLIST CAPTION
        align (top|bottom) #IMPLIED
        >

<!ATTLIST tr                            -- table row --
        %cell.halign;                   -- horizontal alignment in
cells --
        >

<!ATTLIST (th|td)                               -- header or data cell -
-
        rowspan       NUMBER   1        -- number of rows
spanned by cell --
        colspan       NUMBER   1        -- number of cols
spanned by cell --
        %cell.halign;                   -- horizontal alignment
in cell --
        width         %Pixels  #IMPLIED  -- suggested width for
cell --
        >


<!--================ Palm Special
=========================================-->
<!ELEMENT SMALLSCREENIGNORE - - %body.content>


<!--================ Document Head
=========================================-->
```

```
<!-- %head.misc defined earlier on as "META" -->

<!ENTITY % head.content "TITLE & BASE?">

<!ELEMENT HEAD O O  (%head.content) +(%head.misc)>

<!ELEMENT TITLE - -  (#PCDATA)* -(%head.misc)
          -- The TITLE element is not considered part of the flow
of text.
             It is displayed in the menu-bar as the screen title.
           -->

<!--
    The BASE element gives an absolute URL for dereferencing
relative
    URLs, e.g.

         <BASE href="http://foo.com/">
         ...
         <IMG SRC="images/bar.gif">

    The image is deferenced to

         http://foo.com/images/bar.gif

   In the absence of a BASE element the document URL should be
used.
   Note that this is not necessarily the same as the URL used to
   request the document, as the base URL may be overridden by an
HTTP
   header accompanying the document.
-->

<!ELEMENT BASE - O EMPTY>
<!ATTLIST BASE
        href %URL  #REQUIRED
        >

<!--
```

```
Palm OS special name/content pairs:
        name="PalmComputingPlatform" content="true"
             - identifies as Palm Friendly and allows images to
be included

        name="HistoryListText" content="some string"
             - used by clippings to show significant info in
History List

        name="PalmLauncherRevision" content="version"
             - used to provide a version number associated with
the PQA
               to be shown in the launcher info version screen

        name="LocalIcon" content="filename"
             - used to include images or html pages into the PQA
during a
              build when those images or pages aren't referenced
by any of
               the local pages, thus allowing clippings to use
them
-->

<!ELEMENT META - O EMPTY -- Generic Metainformation -->
<!ATTLIST META
        http-equiv  NAME    #IMPLIED  -- HTTP response header
name  --
       name       NAME    #IMPLIED  -- metainformation name
--
        content     CDATA   #REQUIRED -- associated information
                                         (special strings &date
and &time
                                          can be used to include
the date
                                        and time)
--
        >
```

```
<!--================ Document Structure
====================================-->

<!ENTITY % version.attr "VERSION CDATA #FIXED '%HTML.Version;'">

<![ %HTML.Deprecated [
    <!ENTITY % html.content "HEAD, BODY, PLAINTEXT?">
]]>

<!ENTITY % html.content "HEAD, BODY">

<!ELEMENT HTML O O  (%html.content)>
<!ATTLIST HTML
        %version.attr;
        >


<!--================ Character Entities for ISO Latin-1
===================-->

<!-- (C) International Organization for Standardization 1986
     Permission to copy in any form is granted for use with
     conforming SGML systems and applications as defined in
     ISO 8879, provided this notice is included in all copies.
     This has been extended for use with HTML to cover the full
     set of codes in the range 160-255 decimal.
-->
<!-- Character entity set. Typical invocation:
     <!ENTITY % ISOlat1 PUBLIC
       "ISO 8879-1986//ENTITIES Added Latin 1//EN//HTML">
     %ISOlat1;
-->
    <!ENTITY nbsp   CDATA " " -- no-break space -->
    <!ENTITY iexcl  CDATA "&#161;" -- inverted exclamation mark -
->
    <!ENTITY cent   CDATA "&#162;" -- cent sign -->
    <!ENTITY pound  CDATA "&#163;" -- pound sterling sign -->
    <!ENTITY curren CDATA "&#164;" -- general currency sign -->
    <!ENTITY yen    CDATA "&#165;" -- yen sign -->
    <!ENTITY brvbar CDATA "&#166;" -- broken (vertical) bar -->
```

```
    <!ENTITY sect   CDATA "&#167;" -- section sign -->
    <!ENTITY uml    CDATA "&#168;" -- umlaut (dieresis) -->
    <!ENTITY copy   CDATA "&#169;" -- copyright sign -->
    <!ENTITY ordf   CDATA "&#170;" -- ordinal indicator, feminine
-->
    <!ENTITY laquo  CDATA "&#171;" -- angle quotation mark, left
-->
    <!ENTITY not    CDATA "&#172;" -- not sign -->
    <!ENTITY shy    CDATA "&#173;" -- soft hyphen -->
    <!ENTITY reg    CDATA "&#174;" -- registered sign -->
    <!ENTITY macr   CDATA "&#175;" -- macron -->
    <!ENTITY deg    CDATA "&#176;" -- degree sign -->
    <!ENTITY plusmn CDATA "&#177;" -- plus-or-minus sign -->
    <!ENTITY sup2   CDATA "&#178;" -- superscript two -->
    <!ENTITY sup3   CDATA "&#179;" -- superscript three -->
    <!ENTITY acute  CDATA "&#180;" -- acute accent -->
    <!ENTITY micro  CDATA "&#181;" -- micro sign -->
    <!ENTITY para   CDATA "&#182;" -- pilcrow (paragraph sign) --
>
    <!ENTITY middot CDATA "&#183;" -- middle dot -->
    <!ENTITY cedil  CDATA "&#184;" -- cedilla -->
    <!ENTITY sup1   CDATA "&#185;" -- superscript one -->
    <!ENTITY ordm   CDATA "&#186;" -- ordinal indicator,
masculine -->
    <!ENTITY raquo  CDATA "&#187;" -- angle quotation mark, right
-->
    <!ENTITY frac14 CDATA "&#188;" -- fraction one-quarter -->
    <!ENTITY frac12 CDATA "&#189;" -- fraction one-half -->
    <!ENTITY frac34 CDATA "&#190;" -- fraction three-quarters -->
    <!ENTITY iquest CDATA "&#191;" -- inverted question mark -->
    <!ENTITY Agrave CDATA "&#192;" -- capital A, grave accent -->
    <!ENTITY Aacute CDATA "&#193;" -- capital A, acute accent -->
    <!ENTITY Acirc  CDATA "&#194;" -- capital A, circumflex
accent -->
    <!ENTITY Atilde CDATA "&#195;" -- capital A, tilde -->
    <!ENTITY Auml   CDATA "&#196;" -- capital A, dieresis or
umlaut mark -->
    <!ENTITY Aring  CDATA "&#197;" -- capital A, ring -->
    <!ENTITY AElig  CDATA "&#198;" -- capital AE diphthong
(ligature) -->
```

```
    <!ENTITY Ccedil CDATA "&#199;" -- capital C, cedilla -->
    <!ENTITY Egrave CDATA "&#200;" -- capital E, grave accent -->
    <!ENTITY Eacute CDATA "&#201;" -- capital E, acute accent -->
    <!ENTITY Ecirc  CDATA "&#202;" -- capital E, circumflex
accent -->
    <!ENTITY Euml   CDATA "&#203;" -- capital E, dieresis or
umlaut mark -->
    <!ENTITY Igrave CDATA "&#204;" -- capital I, grave accent -->
    <!ENTITY Iacute CDATA "&#205;" -- capital I, acute accent -->
    <!ENTITY Icirc  CDATA "&#206;" -- capital I, circumflex
accent -->
    <!ENTITY Iuml   CDATA "&#207;" -- capital I, dieresis or
umlaut mark -->
    <!ENTITY ETH    CDATA "&#208;" -- capital Eth, Icelandic -->
    <!ENTITY Ntilde CDATA "&#209;" -- capital N, tilde -->
    <!ENTITY Ograve CDATA "&#210;" -- capital O, grave accent -->
    <!ENTITY Oacute CDATA "&#211;" -- capital O, acute accent -->
    <!ENTITY Ocirc  CDATA "&#212;" -- capital O, circumflex
accent -->
    <!ENTITY Otilde CDATA "&#213;" -- capital O, tilde -->
    <!ENTITY Ouml   CDATA "&#214;" -- capital O, dieresis or
umlaut mark -->
    <!ENTITY times  CDATA "&#215;" -- multiply sign -->
    <!ENTITY Oslash CDATA "&#216;" -- capital O, slash -->
    <!ENTITY Ugrave CDATA "&#217;" -- capital U, grave accent -->
    <!ENTITY Uacute CDATA "&#218;" -- capital U, acute accent -->
    <!ENTITY Ucirc  CDATA "&#219;" -- capital U, circumflex
accent -->
    <!ENTITY Uuml   CDATA "&#220;" -- capital U, dieresis or
umlaut mark -->
    <!ENTITY Yacute CDATA "&#221;" -- capital Y, acute accent -->
    <!ENTITY THORN  CDATA "&#222;" -- capital THORN, Icelandic --
>
    <!ENTITY szlig  CDATA "&#223;" -- small sharp s, German (sz
ligature) -->
    <!ENTITY agrave CDATA "&#224;" -- small a, grave accent -->
    <!ENTITY aacute CDATA "&#225;" -- small a, acute accent -->
    <!ENTITY acirc  CDATA "&#226;" -- small a, circumflex accent
-->
    <!ENTITY atilde CDATA "&#227;" -- small a, tilde -->
```

```
    <!ENTITY auml   CDATA "&#228;" -- small a, dieresis or umlaut
mark -->
    <!ENTITY aring  CDATA "&#229;" -- small a, ring -->
    <!ENTITY aelig  CDATA "&#230;" -- small ae diphthong
(ligature) -->
    <!ENTITY ccedil CDATA "&#231;" -- small c, cedilla -->
    <!ENTITY egrave CDATA "&#232;" -- small e, grave accent -->
    <!ENTITY eacute CDATA "&#233;" -- small e, acute accent -->
    <!ENTITY ecirc  CDATA "&#234;" -- small e, circumflex accent
-->
    <!ENTITY euml   CDATA "&#235;" -- small e, dieresis or umlaut
mark -->
    <!ENTITY igrave CDATA "&#236;" -- small i, grave accent -->
    <!ENTITY iacute CDATA "&#237;" -- small i, acute accent -->
    <!ENTITY icirc  CDATA "&#238;" -- small i, circumflex accent
-->
    <!ENTITY iuml   CDATA "&#239;" -- small i, dieresis or umlaut
mark -->
    <!ENTITY eth    CDATA "&#240;" -- small eth, Icelandic -->
    <!ENTITY ntilde CDATA "&#241;" -- small n, tilde -->
    <!ENTITY ograve CDATA "&#242;" -- small o, grave accent -->
    <!ENTITY oacute CDATA "&#243;" -- small o, acute accent -->
    <!ENTITY ocirc  CDATA "&#244;" -- small o, circumflex accent
-->
    <!ENTITY otilde CDATA "&#245;" -- small o, tilde -->
    <!ENTITY ouml   CDATA "&#246;" -- small o, dieresis or umlaut
mark -->
    <!ENTITY divide CDATA "&#247;" -- divide sign -->
    <!ENTITY oslash CDATA "&#248;" -- small o, slash -->
    <!ENTITY ugrave CDATA "&#249;" -- small u, grave accent -->
    <!ENTITY uacute CDATA "&#250;" -- small u, acute accent -->
    <!ENTITY ucirc  CDATA "&#251;" -- small u, circumflex accent
-->
    <!ENTITY uuml   CDATA "&#252;" -- small u, dieresis or umlaut
mark -->
    <!ENTITY yacute CDATA "&#253;" -- small y, acute accent -->
    <!ENTITY thorn  CDATA "&#254;" -- small thorn, Icelandic -->
    <!ENTITY yuml   CDATA "&#255;" -- small y, dieresis or umlaut
mark -->
```

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